Cost & Management Accounting

CA Intermediate — Short Revision Notes

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Contents

Introduction to Cost & Management Accounting

Snapshot

Cost accounting attaches cost to the right cost object (product/job/process/department) at the right time in the right form for a specific decision. Financial accounting looks backward at the whole entity; cost & management accounting look at parts, forward, for insiders. Core skill = choosing the right cost classification ("lens") for the question asked. Same rupee wears different labels for different decisions.

Core concepts

Three information systems:

System	Audience	Time	Governed by
Financial accounting	External (shareholders, lenders, tax)	Past	Companies Act 2013, AS/Ind AS
Cost accounting	Internal + statutory where notified	Past & future	ICAI Cost Accounting Standards; Sec 148 cost records/audit
Management accounting	Internal managers only	Future	No external rulebook

- Cost accounting is partly regulated (cost records/cost audit under **Sec 148, Companies Act 2013** for notified industries/thresholds); management accounting has no statutory format, no audit, no fixed periodicity, no double-entry.
- Cost accounting = ascertains & controls cost (subset/raw material). Management accounting = uses cost + financial + non-financial data to advise (finished advice).
- Nesting: **Costing ⊂ Cost Accounting ⊂ Cost Accountancy**; Management Accounting is outermost.
- CAS-1 definition of cost: "a measurement, in monetary terms, of the amount of resources used for production of goods or rendering of services."

Key provisions / rules

Elements of cost: Material, Labour, Expenses; each splits by traceability into Direct/Indirect. Overhead = Indirect Material + Indirect Labour + Indirect Expenses, grouped by function (Factory/Works, Administration, Selling & Distribution).

Cost build-up:

Stage	Formula
Prime Cost	Direct Material + Direct Labour + Direct Expenses
Works/Factory Cost	Prime Cost + Factory OH (± opening/closing WIP)
Cost of Production	Works Cost + Admin OH
Cost of Goods Sold	Cost of Production + Opening FG − Closing FG
Cost of Sales	COGS + Selling & Distribution OH
Sales	Cost of Sales + Profit
Contribution	Sales − Variable Cost
Material consumed	Opening RM + Purchases − Closing RM

Direct expenses (chargeable expenses): non-material, non-labour costs traceable to one unit (job-specific tool hire, design royalty per unit, special drawing for a contract).

Classifications (each lens serves a decision):

Classification	Split	Decision served
Element	Material/Labour/Expenses	Cost build-up
Function	Production/Admin/Selling-Distribution	Responsibility grouping
Traceability	Direct/Indirect	Product costing; overhead absorption
Behaviour	Fixed/Variable/Semi-variable (incl. step)	Pricing, break-even, extra-order, shutdown
Period-attach	Product/Period	Inventory valuation; timing of profit
Controllability	Controllable/Uncontrollable	Responsibility accounting; appraisal
Normality	Normal/Abnormal	Whether it enters cost at all
Relevance	Relevant/Sunk/Opportunity/Differential/Marginal	Make-or-buy, order, replace, shutdown

- **Direct vs Indirect** decided by convenience/economy of tracing, NOT physical importance (nails/glue = indirect). Cost object must be fixed first — foreman salary is indirect to one cycle but direct to the whole line.
- **Behaviour:** Variable — total ∝ activity, per-unit constant. Fixed — total constant within relevant range, per-unit falls with volume. Semi-variable — fixed base + variable slope (split via high-low). **Fixed cost per unit is variable; variable cost per unit is fixed.** Step (stepped) fixed cost: constant over a range, jumps, constant again. Committed fixed (unavoidable short-run: rent, depreciation) vs Discretionary/managed (management-set: advertising, R&D — first to cut).
- High-low: Variable rate = (Cost at high − Cost at low) ÷ (High units − Low units); fixed = balancing figure.
- **Product cost** = inventoriable (DM+DL+factory OH), sits in stock until sold. **Period cost** = expensed by calendar (selling, distribution, admin; under marginal costing ALL fixed OH). When closing stock rises, absorption profit > marginal profit.
- **Controllable** relative to a manager level AND time horizon; judge managers only on what they control.
- **Normal in, abnormal out:** normal loss/idle time absorbed into product cost; abnormal excluded → Costing P&L.
- **Relevance test:** a cost is relevant only if it is a FUTURE cost that DIFFERS between alternatives.
    - Sunk = already incurred, unrecoverable → ignore.
    - Opportunity = value of next-best alternative forgone → relevant even though no cash moves.
    - Differential/incremental = total cost gap between two courses of action (can include fixed-cost changes).
    - Marginal = cost of one more unit ≈ variable cost per unit.
    - Avoidable (disappears if activity dropped) vs unavoidable.
    - Imputed/notional (notional rent/interest — no cash); Replacement cost (current market); Out-of-pocket (actual cash, excludes depreciation).
    - Explicit (cash, recorded) vs Implicit (imputed/notional/opportunity, unrecorded).

Cost centres & units: Personal vs Impersonal; Production vs Service (service re-apportioned to production). Responsibility centres: Cost / Revenue / Profit / Investment (ROI/RI). *Centre absorbs cost; unit carries cost.* Composite cost units: tonne-km (goods transport), passenger-km, patient-day/bed-day (hospital), room-day (hotel), kWh (electricity), per 1,000 bricks.

Methods (fixed by product) vs Techniques (chosen by decision):

- Methods: Job, Batch, Contract, Process, Operating/Service, Single/Output, Multiple/Composite. Root = specific-order vs continuous-operation.
- Techniques: Absorption, Marginal, Standard, Budgetary Control, Uniform, Historical. Time axis: historical (after) vs predetermined (before).

Cost control vs cost reduction: control keeps cost within standard (standard assumed correct); reduction permanently lowers cost below standard (standard itself challengeable, continuous, never ends).

Worked mini-example

Special order: full cost per cycle ₹3,200 but a dealer offers ₹4,200 for 500 extra units, spare capacity, no S&D. Relevant (variable) cost = DM 1,500 + DL 600 + DExp 150 = ₹2,250. Incremental contribution = (4,200 − 2,250) × 500 = **₹9,75,000 →**

ACCEPT. Fixed factory/admin/S&D (₹950) irrelevant. If offer were ₹2,800 (below full cost ₹3,200 but above ₹2,250), naive rejection loses ₹550 × 500 = ₹2,75,000. If no spare capacity, add opportunity cost of displaced sales.

Exam traps & must-remember

- "Cost per unit" at a new volume: strip out fixed portion first (fixed cost per unit is not constant) — the #1 silent trap.
- Only specific/avoidable fixed cost is relevant to shutdown; apportioned common cost continues → irrelevant.
- Never include sunk cost (past R&D, historical machine cost); never omit opportunity cost.
- Direct/indirect by traceability, not physical importance.
- Method vs technique: marginal is a technique, batch is the method.
- Profit "on cost" vs "on sales": on sales → Sales = Cost ÷ 0.80; on cost → Sales = Cost × 1.20.
- Stock adjusts at three levels: raw material before Prime, WIP at Works cost, finished goods after Cost of Production. Direction: + opening, − closing. Value closing FG at CURRENT period cost of production per unit.
- Pure-financial items NEVER in cost sheet: income tax, dividends, donations, interest on loans/debentures, loss/profit on sale of assets, goodwill written off, fines/penalties.
- Composite units: watch return leg / empty running; state absolute (per actual leg) vs commercial (avg load × total distance) tonne-km.
- Shutdown rule: close only when avoidable fixed cost saved + opportunity gains > contribution lost.

One-line recall

- Cost = a number-for-a-purpose; pick the lens the decision needs.
- Relevant cost = future + differs; ignore sunk, add opportunity.
- Method chosen by product; technique chosen by decision; classification chosen by the question.
- Normal loss absorbed by good output; abnormal loss → Costing P&L.
- Centre absorbs cost, unit carries cost; costing itself passes a cost-benefit test.
- Absorption defers fixed OH into stock → higher profit than marginal when stock rises.

Material Cost

Snapshot

Material is the largest cost element (40–70% of total) and the leakiest (portable, deteriorates, ties up working capital). Control = keep the right quantity (levels + EOQ), price issues honestly (FIFO/LIFO/WA), classify losses (normal vs abnormal), and apply selective control (ABC). Chain of custody with a document at every link: requisition → PO → GRN → bin card/stores ledger → requisition note.

Core concepts

- **Cost-flow assumption, not physical flow:** FIFO/LIFO only assume a costing order; physical issue can differ.
- **Landed cost:** material cost = invoice price − trade discount − creditable GST/duty + carriage inward + insurance in transit + other purchase charges. **Cash discount excluded** (financial item); non-creditable GST added.
- **EOQ tug-of-war:** ordering cost falls with big lots, carrying cost rises with big lots; minimum at the crossing (ordering = carrying). Total-cost curve flat near bottom (robust to ±20–30% error).
- **Levels** = safety-buffer system letting a storekeeper run day-to-day control; management sets levels once.

Key provisions / rules

Procurement documents:

Document	Records	Kept by
Purchase Requisition	Authorised need	Stores/Dept
Purchase Order	Commitment to supplier	Purchasing
Goods Received Note (GRN)	Arrival + inspection	Receiving
Bin Card	Quantity only	Storekeeper (in stores)
Stores Ledger	Quantity + value	Costing office
Material Requisition Note	Authorised issue	Production → Stores

Perpetual inventory (continuous records) vs continuous stocktaking (physical counting that verifies records) — don't confuse.

Inventory levels:

Item	Formula
Re-order Level (ROL)	Maximum usage × Maximum lead time
ROL (alt)	Safety stock + (Avg usage × Avg lead time)
Minimum Level	ROL − (Average usage × Average lead time)
Maximum Level	ROL + ROQ − (Minimum usage × Minimum lead time)
Danger Level	Average usage × Emergency lead time
Average Stock	Minimum level + ½ ROQ = ½(Min + Max)
Safety stock	(Max usage × Max LT) − (Avg usage × Avg LT) = Max·Max − Avg·Avg

Ordering must hold: Danger < Minimum < Re-order < Maximum.

EOQ:

- EOQ = √(2 × A × O ÷ C)
- A = annual demand (units); O = ordering cost per order; C = carrying cost per unit per year = carrying % × unit price.

- No. of orders/yr = A ÷ EOQ; Time between orders = 365 ÷ (A/EOQ) days.
- Ordering cost = (A/EOQ)×O; Carrying cost = (EOQ/2)×C; **at EOQ these are equal (self-check).**
- Square-root damping: demand ×4 → EOQ ×2; ordering cost ×2 → EOQ ×√2; carrying cost ×2 → EOQ ÷√2.
- Quantity discounts: minimise Total = (A × price) + (A/Q)×O + (Q/2)×C. Evaluate at plain EOQ and at smallest qualifying quantity of each slab; recompute C at each slab's price. Deepest discount need not win.
- Assumptions: known constant demand; constant O and C; constant price (no discounts); whole order delivered at once; known lead time; no stock-outs.

Issue valuation (rising prices):

Method	Issue cost	Profit	Closing stock
FIFO (oldest issued first)	Lowest	Highest	Highest
LIFO (newest first; banned under AS-2/Ind AS-2)	Highest	Lowest	Lowest
Weighted Average	Middle	Middle	Middle

- Weighted-avg rate = value on hand ÷ units on hand, recomputed after each **receipt** (issue doesn't change rate). Periodic weighted average = one rate for whole period.
- Notional-price methods (standard, replacement/market) throw up a variance → Costing P&L; actual-price methods recover exactly what was paid.
- Simple average (average of rates, ignores quantities) can break reconciliation — theoretically unsound.
- Universal check: Opening + Purchases = Issues + Closing (in quantity AND value). Falling prices → reverse FIFO/LIFO.

Losses (golden principle: normal absorbed by good output; abnormal → Costing P&L):

Type	Recovery	Treatment
Waste (no recovery)	Nil/negative	Normal → good output; Abnormal → Costing P&L
Scrap (small sale value)	Yes small	Normal scrap credited to cost/OH; Abnormal → Costing P&L
Spoilage (unrectifiable)	Low	Normal net cost → good output; Abnormal net → Costing P&L
Defectives (rectifiable)	Rectifiable	Normal rectification → good output/OH; Abnormal → Costing P&L

- Cost per good unit = (Total cost − normal-loss scrap value) ÷ **expected good output** (input − normal loss).
- Abnormal loss = expected good output − actual output; value at cost per good unit, less its own scrap → Costing P&L.
- Abnormal gain = actual output − expected good output; value at same rate; DEBIT process / CREDIT Costing P&L; then claw back scrap on the normal loss that did NOT occur (reduces the gain).

Selective control:

Class	~% items	~% value	Control
A	10%	70%	Tight, low safety stock, frequent review, EOQ enforced
B	20%	20%	Moderate
C	70%	10%	Loose, bulk buy, high safety stock, two-bin

Classify by **annual consumption value = annual qty × unit price** (not sticker price). Cousins: VED (criticality — spares), FSN (movement — dead stock), HML (unit price — authorisation), SDE (availability — imports).

Inventory turnover = Cost of material consumed ÷ Average stock [Avg = ½(Opening+Closing)]; Holding days = 365 ÷ turnover. Low/falling turnover = slow-moving/obsolete → investigate.

Worked mini-example

A = 12,000 units/yr, O = ₹150/order, unit price ₹20, carrying 20% → C = ₹4. EOQ = √(2 × 12,000 × 150 ÷ 4) = √9,00,000 = **948.68 ≈ 949 units.** Orders/yr = 12,000 ÷ 948.68 = 12.65 (~13). Time between = 365 ÷ 12.65 ≈ 29 days. Ordering cost = 12.65 × 150 = ₹1,897; Carrying = (948.68/2) × 4 = ₹1,897 (equal → correct). Total inventory cost ≈ ₹3,795.

Exam traps & must-remember

- Carrying cost given as % → convert to ₹ per unit per year (% × unit price) before EOQ. #1 error.
- O is per order, not annual purchasing-dept cost.
- ROL uses MAX × MAX; Minimum subtracts AVG × AVG; Maximum subtracts MIN × MIN. Keep usage and lead time on the same time basis (don't mix days/weeks).
- Quantity discounts: compare total annual cost (purchase + ordering + carrying) at EOQ and each break-point; deepest discount need not win.
- Cost per good unit uses EXPECTED good output, not actual output.
- Scrap routing: normal-loss scrap reduces cost-per-unit; abnormal-loss scrap nets against Costing P&L; abnormal gain claws back scrap on loss that didn't occur.
- Weighted average recomputed after every RECEIPT (not period-end, not on issues).
- FIFO = higher profit only in RISING prices; reverse if falling.
- ABC by annual consumption value, not unit price; ABC ignores criticality (use VED for vital cheap spares).
- Cash discount excluded from material cost; creditable GST excluded; carriage inward added.
- Bin card = quantity only (storekeeper); stores ledger = quantity + value (costing office).

One-line recall

- EOQ = √(2AO/C); at EOQ ordering cost = carrying cost.
- ROL = Max usage × Max lead time; Danger = Avg usage × Emergency lead time.
- Rising prices: FIFO → highest profit & closing stock; LIFO → lowest; WA middle.
- Normal loss absorbed by good output; abnormal loss/gain → Costing P&L at cost-per-good-unit rate.
- ABC by consumption value; classify attention where rupees are.
- Check: Opening + Purchases = Issues + Closing (qty & value).

Employee (Labour) Cost

Snapshot

Labour is paid for TIME but valued for OUTPUT; the gap (idle time, inefficiency) is where cost leaks. Employee cost = wages + DA + bonus + PF/ESI + gratuity + leave pay + recruitment/training. Direct = traceable to a cost object; indirect = supervision, general idle time, fringe benefits. Core tasks: classify idle time/overtime, measure turnover, and design incentive pay that aligns worker's gain with firm's profit.

Core concepts

Master decision rule for any labour cost item:

- Caused by a specific job → DIRECT (charge to job).
- General cause, normal/unavoidable → Factory OVERHEAD (spread over all jobs).
- Abnormal/avoidable → Costing P&L (spotlight the loss).

Key principle (firm vs worker time-frames): a faster worker earns MORE per hour (bonus) but the firm's cost PER UNIT falls (fixed overhead spread over more units in fewer hours). "Which plan is best for the firm?" = compare cost per unit, not earnings.

- **Time-keeping** = was the worker present, how long? (attendance → payment; register, disc/token, clock, biometric).
- **Time-booking** = what did the worker do inside? (job card/time sheet → cost allocation).
- Idle time discovered by reconciling the two: **Idle Time = Time paid (attendance) – Time booked to jobs.**

Key provisions / rules

Idle time treatment:

Type	Cause	Treatment
Normal — job-related	Setup, waiting between operations for a specific job	Charge to the job (inflate labour rate)
Normal — general	Tea breaks, small unavoidable waits, flow-line balancing, seasonal slack	Factory overhead
Abnormal	Strike, lockout, power failure, breakdown, flood, poor planning	Costing P&L

Effective (inflated) rate = Total wage ÷ Productive (effective) hours. Pull abnormal idle wages OUT to P&L first, then inflate over remaining productive hours.

Overtime (Factories Act 1948 — typically 2× ordinary rate; verify current):

- Ordinary wage for OT hours → always product cost.
- OT premium (extra over normal) treated by cause: customer rush → that job (direct); general pressure → factory overhead; abnormal (breakdown make-up) → Costing P&L; worker's fault/redoing defectives → worker/dept or P&L.
- OT premium per OT hour (double pay) = 2R – R = R.
- Idle time = too many hours for the work; overtime = too little. A breakdown can create idle time now + overtime later — both go to Costing P&L.

Labour turnover (× 100, over Average workers = (Opening+Closing)/2):

Method	Formula
Separation	Separations ÷ Avg workers

Method	Formula
Replacement	Replacements only ÷ Avg workers
Flux (accessions form)	(Separations + Accessions) ÷ Avg workers
Flux (replacement form)	(Separations + Replacements) ÷ Avg workers
Annualised (short period)	(Period LTR ÷ days) × 365

- Separations = all exits (resign, discharge, retrench, retire, death). Replacements = hires to fill vacancies (EXCLUDES expansion). Accessions = all hires = replacements + expansion.
- Costs: Preventive (welfare, conditions, retention — spread over whole workforce) vs Replacement (recruitment, training, lost output, spoilage — attach to churn departments). Optimum ≠ zero turnover; minimise preventive + replacement combined.

Remuneration systems (risk dial: time rate = all risk on firm; piece rate = all risk on worker; premium plans share saved time):

Plan	Signature	Floor?
Time rate	Hours × rate; zero incentive, protects quality	Yes (is the floor)
Straight piece rate	Units × rate; strong incentive, quality/wastage risk, no min wage	No
Taylor differential	Low rate below std (~83%), high above (~125%); no minimum, punishing	No
Merrick	3 bands, never below ordinary piece rate: ≤83% ordinary, 83–100% →110%, >100% →120%	Ordinary rate
Halsey	Time wage + 50% of saved-time value	Yes
Rowan	Time wage + (saved ÷ standard) × time wage; self-caps at 50%	Yes
Gantt task & bonus	Guaranteed wage; +20% bonus at standard; high piece rate (120%) above; a cliff/step	Yes
Emerson efficiency	No bonus below ~66⅔%; graduated bonus to ~20% at 100%; +1%/1% above — smooth ramp	Yes
Bedaux (point)	Work in "B" units; 75% of saved time to worker, 25% to foreman	Yes
Barth	R × √(std hours × hours worked); for learners, no minimum	No
Group bonus	Team earns, shared by time-wages / equally / agreed %	Per member

Key formulas (S = standard time, T = time taken, R = rate/hr; time saved = S − T):

- **Halsey earnings = (T × R) + 50% × (S − T) × R** (Halsey-Weir variant uses 30–33⅓%).
- **Rowan earnings = (T × R) + [(S − T) ÷ S] × T × R.** Rowan bonus maximised at 50% saved; **never exceeds the time wage (T × R).**
- Labour efficiency = (Std time for actual output ÷ Actual time) × 100.
- Effective rate/hour = Total earnings ÷ Actual hours (T).
- Cost per unit to firm = (Wages + Hours × OH rate) ÷ Units produced.

Halsey vs Rowan crossover (at T = ½S, i.e. 50% time saved they are EQUAL):

- Below 50% saved → Rowan pays more (generous to modest improvers).
- Above 50% saved → Halsey pays more (rewards the star; Rowan's ceiling holds it back).

Worked mini-example

S = 60 hrs, T = 40 hrs, R = ₹15/hr, time saved = 20 hrs (33.3% of standard).

- Time rate = 40 × 15 = ₹600 (eff ₹15/hr).
- Halsey = 600 + 0.5 × 20 × 15 = 600 + 150 = ₹750 (eff ₹18.75/hr).
- Rowan = 600 + (20/60) × 40 × 15 = 600 + 200 = ₹800 (eff ₹20/hr).
- Piece rate (job worth 60 std hrs) = 60 × 15 = ₹900 (eff ₹22.50/hr). Time saved 33.3% < 50% → Rowan (₹800) > Halsey (₹750), as theory predicts.

Exam traps & must-remember

- Replacement rate uses ONLY replacements; expansion hires are accessions, not replacements. #1 turnover trap.
- Only the cause-appropriate portion of the OT premium is traced; ordinary-rate element on OT hours is always product cost.
- Under Halsey/Rowan, if T > S there is NO negative bonus — worker still gets T × R (guaranteed floor).
- Rowan bonus multiplies by T (time taken), not S.
- Crossover direction: below 50% saved Rowan > Halsey; above 50% Halsey > Rowan; equal at 50%.
- Abnormal idle time → Costing P&L, never into job/overhead; strip abnormal wages out before inflating the rate.
- Effective (inflated) rate is HIGHER than paid rate (same money over fewer hours).
- Turnover denominator = average workers, not closing.
- When overhead per hour is given, compute cost per unit/job TO THE FIRM — a plan paying more per hour can be cheaper per unit (fewer hours).
- Taylor: no minimum, punishes slow. Merrick: never below ordinary piece rate. Gantt = step/cliff; Emerson = smooth ramp.
- Group bonus: "in proportion to wages" ≠ "equally" — show the sharing ratio line.
- Verify named percentages (Taylor 83/125, Merrick bands, Emerson 66⅔, Bedaux 75/25) against current ICAI material.

One-line recall

- Idle time = time paid − time booked; reconcile time-keeping vs time-booking to expose it.
- Cause decides charge: job → direct; general → overhead; abnormal → Costing P&L.
- Halsey = Half of saved time; Rowan = Ratio (saved/standard) × time wage, self-caps at 50%.
- Below 50% saved Rowan wins; above 50% Halsey wins; equal at 50%.
- Replacement excludes expansion; separations include every exit.
- Worker counts rupees per hour; firm counts rupees per unit.

Overheads (Absorption Costing)

Snapshot

- Overhead = Indirect Material + Indirect Labour + Indirect Expenses — real cost that points at no single product.
- "Indirect" is relative: an item is overhead when tracing to the unit is impossible (rent) or not economically worthwhile (dab of glue) — the convenience & materiality test.
- Core flow: Total OH → (Allocation + Apportionment = Primary distribution) → cost centres → (Re-apportionment = Secondary distribution) → production depts → (Absorption) → units.
- Three destinations: Factory OH → product cost & closing stock; Admin OH → generally period cost recovered on cost of production; S&D OH → recovered only on goods sold, never in stock.

Core concepts

- **Allocation** = charge whole item to one centre (100% traceable). **Apportionment** = split a shared item across centres on a fair basis. **Absorption** = charge production-dept OH onto units via a rate. Allocation + apportionment = primary distribution; re-apportionment of service depts = secondary distribution.
- **Collection** (accumulate by nature, standing-order/cost-account number) precedes allocation (charge to a centre).
- Allocation vs apportionment razor = traceability of the whole. Separately metered power → allocate; single factory bill → apportion. A line can mix both (e.g. power ₹50,000 with ₹12,000 separately metered to A → allocate ₹12,000, apportion ₹38,000).
- Production depts work on the product (units absorb cost); service depts (Stores, Maintenance, Canteen, Power House) serve other depts, so their cost must first be pushed to production depts.
- **Pre-determined rate** = Budgeted OH ÷ Budgeted (normal) activity — needed because prices are quoted before period ends; the mismatch with actuals causes under/over-absorption (a fixed-OH phenomenon).
- **Normal capacity** anchor: divide fixed OH by normal (long-run average, after normal idle time) capacity so idle-capacity cost is exposed as a period loss, not hidden in units.

Key provisions / rules

Semi-variable split (know all 4): High–Low, Comparison/range, Least-squares regression, Graphical/scatter.

- High–Low: Variable/unit = (Cost_high − Cost_low) ÷ (Units_high − Units_low); Fixed = Total − Variable × Activity.

Apportionment bases (memorise pairing):

Overhead	Basis
Rent, rates, building dep./repairs	Floor area (sq ft)
Lighting, heating	Floor area / light points (use light points/wattage if given); heating on volume if given
Power / electricity	HP × machine hours (or metered KWH)
Depreciation & insurance of plant	Value / capital of plant
Supervision, canteen, welfare, ESI, PF, time-keeping	Number of employees
Stores / material handling	Value or weight of material issued
Fire insurance of stock	Average value of stock held
Delivery / distribution	Weight, volume, tonne-km
General / indirect wages	Direct wages or direct labour hours

Rule: always pick the **most cause-specific basis the data permits**.

Primary distribution cross-checks: (1) each row sums across depts to item total; (2) primary-total row = grand total.

Secondary distribution method chooser:

Situation	Method
Ignore inter-service work	Direct re-distribution (re-base % onto production depts only, sum to 100%)
One-way service between service depts	Step-ladder — rank by number of depts served; larger overhead breaks ties; once closed nothing returns
Mutual service, exactly 2 service depts	Simultaneous equations
Mutual service, 2+ service depts	Repeated distribution (iterate till service balances ≈ 0)

Simultaneous-equation template: S1 = a + p·S2 ; S2 = b + q·S1 → solve → distribute *solved totals* to production depts in original %. Reciprocal methods must agree (differ only by rounding). Extends to 3 depts via 3 equations.

Absorption rate (general) = Production dept overhead ÷ Total base quantity. Six bases:

Method	Formula	Use when
% of Direct Material	(Prod OH ÷ Direct Material) × 100	rarely fair (price swings)
% of Direct Wages	(Prod OH ÷ Direct Wages) × 100	uniform labour rates
% of Prime Cost	(Prod OH ÷ Prime Cost) × 100	seldom ideal
Labour Hour Rate	Prod OH ÷ Direct Labour Hours	**labour-intensive**
Machine Hour Rate	Prod OH ÷ Machine Hours	**machine-intensive**
Rate per unit	Prod OH ÷ Units produced	homogeneous output

Time-based rates (labour hr, machine hr) preferred — overhead is fundamentally time-related; money bases get distorted by price fluctuation.

Machine Hour Rate: MHR = Standing charges/hr + Machine running expenses/hr. Denominator = **effective (productive) hours** = gross − normal idle (setup, maintenance, breakdowns). Abnormal idle is NOT deducted (→ Costing P&L). Depreciation for MHR = (Cost − Residual) ÷ Life, then ÷ effective hours. One operator/one machine → wage may be direct (out of MHR); one operator/several machines → indirect (inside comprehensive MHR). "Comprehensive MHR" usually includes wages + setup.

Blanket rate = total factory OH ÷ total factory base (one rate). Acceptable only when single product OR all products pass uniformly through all departments; else use **departmental rates**.

Under/Over-absorption:

- Overhead absorbed = Pre-determined rate × **Actual** base.
- Under/Over = Absorbed − Actual overhead.
- Absorbed > Actual → **over-absorbed** (credit Costing P&L, costing profit understated).
- Absorbed < Actual → **under-absorbed** (debit Costing P&L, costing profit overstated).
- Two causes: Expenditure part = Budgeted OH − Actual OH; Volume part = Std rate × (Actual activity − Budgeted activity). They sum to the total gap (seed of fixed-OH expenditure & volume variances).

Treatment of under/over-absorption:

Cause / size	Treatment
Abnormal (strike, fire, breakdown)	Costing P&L
Sub-normal capacity (idle-capacity cost)	Costing P&L (do NOT spread)
Normal & small	Costing P&L
Normal & large (wrong rate/volume estimate)	**Supplementary rate** → WIP + FG + COGS

Supplementary rate = under/over amount ÷ actual base. Under → positive rate added; over → negative rate deducted. Apply either per unit/hour, or pro-rata on absorbed-OH value in each of WIP/FG/COGS.

Administration OH — 3 views: (1) apportion between production and S&D; (2) charge to Costing P&L as period cost; (3) ICAI default — recover as % of cost of production/works cost. **S&D OH:** recovered only on units sold, never enters stock; bases = % of works cost, % of selling price, rate per unit sold; distribution costs on weight/volume/tonne-km.

Worked mini-example

Machine hour rate. Machine ₹2,40,000, scrap ₹20,000, life 10 yr; dept rent ₹36,000 (machine = 1/4 area); supervision ₹48,000 (1 of 4 machines); power 5 units/hr @ ₹6; R&M ₹11,000; runs 2,200 effective hrs.

- Depreciation = (2,40,000 − 20,000)/10 = ₹22,000; Rent = 36,000 × 1/4 = ₹9,000; Supervision = 48,000 × 1/4 = ₹12,000; R&M ₹11,000. Standing total = ₹54,000.
- Standing/hr = 54,000 ÷ 2,200 = ₹24.545. Power/hr = 5 × 6 = ₹30.
- **MHR = 24.545 + 30 = ₹54.55/hr.** Check: 54.545 × 2,200 = ₹1,20,000 = 54,000 + 66,000 ✓.
- Comprehensive (operator ₹40/hr) = 54.55 + 40 = ₹94.55/hr.

Exam traps & must-remember

- Overhead absorbed uses **actual** base, NOT budgeted (pre-determined rate uses budgeted; application uses actual).
- Direction: Absorbed > Actual = over (credit P&L); Absorbed < Actual = under (debit P&L).
- Simultaneous method: distribute the **solved** service total (not the residual/primary); include service-to-service % — dropping them wrongly makes it the direct method.
- Step method: start with dept serving most others (larger OH breaks ties); never return cost to closed dept; re-base surviving %.
- Abnormal causes (strike, fire, idle capacity, breakdown) → always Costing P&L, never supplementary rate.
- Supplementary rate must hit **WIP too**, not just FG and COGS.
- **Idle-capacity under-absorption (sub-normal working) → Costing P&L, NOT spread** — exception to "normal + large → supplementary rate". Read the cause.
- Power on HP × machine hours, NOT floor area/headcount. Depreciation on plant value, NOT floor area. Supervision on employees, NOT plant value.
- Allocate any separately-metered/named slice first, then apportion the remainder.
- MHR denominator = effective hours (after normal idle), not gross.
- S&D overhead never enters closing-stock value; include factory OH (admin per policy).
- Direct method: renormalise service-to-production % to 100%.
- Blanket rate justification = "single product OR uniform flow", not "convenient".

One-line recall

- Allocate whole, Apportion shared, Absorb into units; Primary then Secondary then Absorption.
- Pre-determined rate = Budgeted OH ÷ Budgeted (normal) activity; Absorbed = rate × actual base; gap = Absorbed − Actual.
- Machine-intensive → machine hour rate; labour-intensive → labour hour rate; time bases beat money bases.
- Reciprocal (mutual) service → simultaneous equations or repeated distribution; both land whole total on production depts and must agree.
- Normal & large gap → supplementary rate over WIP+FG+COGS; abnormal / idle-capacity / small → Costing P&L.
- MHR = standing charges/hr + running expenses/hr, over effective hours.

Activity-Based Costing (ABC)

Snapshot

- Purpose of ABC = a better **decision** (pricing, product mix, make-or-buy, cost reduction), NOT more accurate financial statements. Total profit is unchanged.
- The disease is **averaging**: spreading one overhead pot over one volume base taxes light users to subsidise heavy users (cross-subsidy).
- Causal chain: **Resources → Activities (cost pools) → Products (via cost drivers).** Cost follows causation.
- Distortion law: traditional costing **over-costs high-volume simple** products and **under-costs low-volume complex** products; ABC reverses both. Distortions are equal & opposite (conservation of overhead).

Core concepts

- **Activity** = task consuming resources (setup, inspect, purchase, move, machine). **Cost pool** = total ₹ of one activity. **Cost driver** = factor causing the activity's cost. **Cost driver rate** = pool ÷ total driver quantity.
- **Resource driver** traces resources INTO a pool (stage 1); **activity driver** traces pool TO products (stage 2). Exams usually give pools pre-totalled (stage 1 done).
- **Cost driver** = cause of cost; **cost object** = thing being costed (product, order, customer, channel).
- **Value-added** vs **non-value-added (NVA)** activities — ABC quantifies rupees in NVA (springboard to ABM). Some NVA are necessary (statutory inspection) and cannot be cut.
- Traditional costing designed when labour was dominant and overhead small; now overhead is 40–70% and product ranges fragmented, so blunt volume allocation swings product costs 30–50%.

Key provisions / rules

Master formulae:

- Cost Driver Rate = Total Cost of Activity Pool ÷ Total Cost Driver Quantity
- Overhead to a Product = Σ (Driver Rate × Driver Units consumed by product) [across all activities]
- Traditional Rate = Total Overhead ÷ Total Volume Base (labour hrs / machine hrs / units)
- Machine-hr driver total = Σ (units × machine hrs per unit) — NOT units alone
- Unit cost = (Prime cost + Overhead absorbed) ÷ units
- Reconciliation: Σ overhead absorbed by all products = Total overhead given
- Cross-subsidy: ₹ over-costed on one product = ₹ under-costed on the other

Two kinds of cost driver (accuracy ladder):

- **Transaction driver** — counts times activity happens (no. of setups, orders, inspections); cheap; use when occurrences uniform.
- **Duration driver** — how long it takes (setup hours, inspection hours); use when occurrences vary widely.
- **Intensity / direct-charge driver** — actual resources charged to a special job; dearest, most accurate; for genuinely special jobs.
- Ladder: transaction (cheapest/least accurate) → duration → intensity (dearest/most accurate).

Driver-choice tests (both required): (1) **causation** — driver movement must genuinely cause pool movement; (2) **measurability** — cheap and objective to count. Perfect-but-unmeasurable is useless; measurable-but-non-causal is worse (looks authoritative while lying).

Cooper's hierarchy of activities (most examinable):

Level	Triggered by	Examples	Driver	Avoidable if you…
Unit-level	each unit	power, machining, consumables/unit	machine hrs, labour hrs, units (volume)	drop one unit
Batch-level	each batch/run	setup, first-article inspection, material moves, purchase order	no. of setups, batches, orders, inspections	drop/merge a batch
Product-level	each product line existing	design, BOM maintenance, special testing, dedicated tooling	no. of products, design hours	drop the product line
Facility-level	factory existing	rent, security, general admin, lighting	none — arbitrary allocation or period cost	close the facility

- Batch- and product-level costs are fixed to volume but variable to complexity; batch cost is step-fixed (steps up per batch). Facility-level is genuinely un-traceable.
- Relevance ladder = only costs at or below the level you change are avoidable. Dropping a product does NOT remove facility-level cost.

Seven-step ABC procedure: identify activities → create cost pool per activity (stage-1) → identify cost driver → compute driver rate → measure each product's driver consumption → charge OH (rate × units, summed) → add prime cost.

Granularity trade-off: too few pools = blanket averaging; too many = costly, cluttered. Merge activities driven by the same driver into one **homogeneous cost pool** (costs move in proportion to one driver). Goal = fewest pools that stay homogeneous.

When ABC pays off: overhead is large share of cost · product/customer range diverse · non-volume overheads dominate (setups, inspections, handling) · pricing must be sharp · frequent high-stakes mix/pricing decisions. **Not worth it:** single product · trivial overhead · all overhead genuinely volume-driven.

Limitations: expensive to install/run · not every cost has a clean driver (residual arbitrariness on facility cost — reduces but never eliminates) · over-refinement breeds complexity · still largely full-absorption/historical (full cost ≠ relevant cost for short-run drop/special-order decisions) · behavioural resistance & driver-count gaming.

Cost object flexibility: ABC can cost a product, order, customer or channel. **ABM:** operational ABM = do things right (make activities cheaper — SMED cuts setup, quality-at-source cuts inspection); strategic ABM = do the right things (change mix/design/price). **ABB:** works the plan backward (output → activities → resources) to set driver-based budget; ABC works actuals forward.

Worked mini-example

Standard (1,00,000 units, prime ₹40) & Premium (10,000 units, prime ₹60). Total OH ₹22,00,000; each uses 0.5 mc hr/unit (Standard 50,000, Premium 5,000).

Traditional rate = 22,00,000 ÷ 55,000 = ₹40/mc hr → ₹20 OH/unit both → Standard ₹60, Premium ₹80.

ABC driver rates: Machining 5,50,000÷55,000 = ₹10/hr; Setups 6,00,000÷120 = ₹5,000; Inspection 4,50,000÷150 = ₹3,000; Purchasing 3,00,000÷200 = ₹1,500; Handling 3,00,000÷150 = ₹2,000.

- Premium (setups 100, inspections 120, orders 160, moves 100, mc-hr 5,000): OH = 50,000 + 5,00,000 + 3,60,000 + 2,40,000 + 2,00,000 = ₹13,50,000 → ₹135/unit.
- Standard OH = ₹8,50,000 → ₹8.50/unit. Reconcile: 8,50,000 + 13,50,000 = 22,00,000 ✓.
- **ABC total cost: Standard ₹48.50, Premium ₹195** (vs traditional ₹60 / ₹80). Cross-subsidy = ₹11.50 × 1,00,000 = ₹115 × 10,000 = ₹11,50,000 (equal & opposite ✓). Premium sold at "40% margin" ₹112 actually loses ₹83/unit.

Exam traps & must-remember

- Facility-level costs are NOT driver-traceable — allocate arbitrarily (and say so) or treat as period cost; never invent a spurious driver.
- Machine-related driver total = units × mc hr/unit, NOT units (commonest slip).

- Always **reconcile** absorbed OH back to total OH before writing the comment.
- Keep **4–6 decimals** in driver rates; round only the final answer.
- Don't confuse the two stages — pools usually pre-totalled (stage 1 done); don't re-split.
- ABC raises low-volume cost only because such products are usually complex — mechanism is complexity, not volume per se.
- Total profit is **identical** under ABC and traditional — only per-product profit differs (conceptual error to say otherwise).
- Never recommend dropping a "loss" product on ABC full cost alone — it includes facility-level cost that won't vanish; use avoidable-cost/contribution analysis (full cost ≠ relevant cost).
- Setup hours (duration) vs number of setups (transaction) — use whichever the question offers.
- Keep total-overhead row and per-unit row physically separate; don't divide by units too early or mix per-unit with totals.
- Use the **stated current** traditional base (labour vs machine hrs) — it's a given fact, not your choice; the base changes the magnitude of distortion but never its direction.
- Charge whole setups by count (₹5,000/setup) — never smooth per unit then re-multiply (reintroduces volume-averaging).
- Lowering an over-costed product's cost is not new profit — total profit unchanged; only pricing room changes.
- More pools ≠ more accuracy beyond homogeneity.
- Driver denominator = whole plant's total driver count, not one product's.

One-line recall

- Cost follows causation: Resources → Activities → Products; rate = pool ÷ total driver, charge = rate × units consumed.
- Traditional over-costs high-volume simple, under-costs low-volume complex; ABC reverses both; distortions equal & opposite (conservation).
- Cooper's hierarchy = unit / batch / product / facility; only unit-level is truly volume-driven; facility-level is un-avoidable except by closing the plant.
- Driver ladder: transaction → duration → intensity; a good driver is causal AND measurable.
- ABC pays off with large overhead + diverse products + non-volume drivers + sharp pricing; total profit never changes, only per-product.
- Full cost (ABC) ≠ relevant cost — use avoidable-cost analysis before dropping a product.

Chapter 06 — Cost Sheet

Snapshot

Cost Sheet = a memorandum statement (NOT a ledger account; no Dr/Cr) showing the build-up of cost element-by-element and stage-by-stage up to profit. Cost accumulates as the product travels: directs (Prime) → factory (Works) → office (Cost of Production) → customer (Cost of Sales) → Profit → Sales. Each stock is injected at the stage where the goods physically rest. Historical version = control/stock valuation; estimated version = tenders/quotations.

Core concepts

- **Direct vs Indirect** = economic traceability, not physical presence (glue in a chair = indirect; hired mould for one job = direct expense).
- Classifications: by **element** (material/labour/expense), by **function** (production/admin/S&D), by behaviour (fixed/variable — marginal costing only), by controllability.
- Four milestones + aliases: Prime Cost (=Basic/First/Flat), Works Cost (=Factory/Manufacturing), Cost of Production (=Office/Gross COP), Cost of Sales (=Total Cost).
- **Stock-at-stage:** RM → before Prime; WIP → at Works; FG → after Cost of Production.
- **Stock valuation:** RM at purchase cost; WIP at works cost (or prime cost if stated); FG at cost of production per unit. Selling cost is NEVER carried in stock.

Key provisions / rules

#	Stage	Formula
1	Material Consumed	Op. RM + Purchases + Carriage inward + duty − Returns/Trade disc − Cl. RM
2	**Prime Cost**	Material Consumed + Direct Wages + Direct Expenses
3	Gross Works Cost	Prime Cost + Factory OH − Scrap sale
4	**Works/Factory Cost**	Gross Works Cost + Op. WIP − Cl. WIP
5	**Cost of Production**	Works Cost + Admin OH (production)
6	**COGS**	Cost of Production + Op. FG − Cl. FG
7	**Cost of Sales**	COGS + S&D OH
8	**Sales**	Cost of Sales + Profit

Margin conversions

- Profit x% on cost → Sales = Cost × (1 + x); on-sales rate = x/(100+x).
- Profit y% on sales → Cost = Sales × (1 − y); on-cost rate = y/(100−y).
- Ready reckoner: 25% on cost = 20% on sales; 33⅓% on cost = 25% on sales; 50% on cost = 33⅓% on sales; 100% on cost = 50% on sales; 10% on cost = 9.09% on sales; 20% on cost = 16.67% on sales.
- "Mark-up" = on cost; "margin" = usually on selling price.

Material cost (CAS-6): purchase price net of trade discount/rebate/duty drawback/refundable taxes (GST credit) + freight inward, insurance in transit, loading, octroi/entry tax, normal transit loss. Exclude cash discount, demurrage/penalties, refundable taxes.

Special items

- Overtime premium: job-specific → direct expense; general factory pressure → factory OH; abnormal → Costing P&L. Only the premium reclassified; normal-rate overtime stays in direct wages.
- Idle time: normal → factory OH; abnormal (strike/power fail) → Costing P&L.
- Scrap sale (normal) → deducted from factory OH/works cost; abnormal loss → Costing P&L.

- By-product NRV → deducted from works cost. Normal spoilage → good units; abnormal → P&L. Rectification of normal defectives → factory OH.
- Primary packing (saleability) → production cost; secondary/transport packing → distribution.
- R&D / quality-control cost attributable to production → added at production stage.

Excluded (financial/appropriation): interest on loans/debentures, income-tax, dividends, transfer to reserves, loss/profit on sale of assets/investments, goodwill/preliminary/discount on shares written off, donations, cash discount (recd/allowed), non-operating incomes (interest/dividend/rent received — NOT deducted from cost), abnormal losses, fines/penalties, recoverable taxes (GST credit). Notional costs (rent/interest/proprietor's salary) included ONLY if instructed.

Per-unit base: Material/Prime/Works/COP → ÷ units produced; COGS/Cost of Sales/Sales/Profit → ÷ units sold. **Units sold = Op. FG units + Units produced − Cl. FG units.** FG stock assumes FIFO unless told (opening at last year's rate, closing at current year's).

Worked mini-example

Produced 20,000 units, sold 18,000 @ ₹300; no opening FG/WIP. RM ₹24,00,000; wages ₹12,00,000; direct exp ₹2,00,000; factory OH ₹9,00,000; admin OH ₹5,00,000; S&D ₹3,60,000.

- Prime = 38,00,000; Works = 47,00,000; Cost of Production = 52,00,000 → per unit ₹260.
- Closing FG = 2,000 × ₹260 = ₹5,20,000 (at COP, not cost of sales).
- COGS = 52,00,000 − 5,20,000 = 46,80,000; Cost of Sales = +3,60,000 = 50,40,000.
- Sales = 54,00,000; Profit = ₹3,60,000. Check: COGS ÷ 18,000 = ₹260 ✓.

Exam traps & must-remember

- Wrong rung for a stock corrupts everything below (RM→Prime, WIP→Works, FG→COP).
- Carriage inward = material cost (before Prime); carriage outward = S&D (Cost of Sales).
- Profit "on cost" ≠ "on sales" — convert deliberately.
- Closing FG valued at cost of production, never cost of sales.
- WIP "at prime cost" → adjust BEFORE factory OH (default = after, at works cost); always name the assumption.
- Financial items sneaked in must be excluded; non-operating incomes NOT deducted from cost.
- Depreciation split by location: plant→factory; office equip→admin; delivery van→S&D.
- Per-unit base switches from produced to sold at the FG line.
- Only irrecoverable taxes/duties form part of material cost; GST input credit excluded.

One-line recall

- Ladder: Prime → Works → COP → COGS → Cost of Sales → Sales; stocks RM/WIP/FG at their own stage.
- Material consumed = Op.RM + Purchases + Carriage-in + duty − returns − Cl.RM.
- FG valued at cost of production per unit; selling cost never sits in stock.
- 25% on cost = 20% on sales; convert with x/(100±x).
- Exclude interest, tax, dividends, loss on asset sale, cash discount, abnormal losses.
- Per-unit ÷ produced above COGS, ÷ sold from COGS down.

Chapter 07 — Cost Accounting Systems (Integrated & Non-Integrated) + Reconciliation

Snapshot

Financial and cost books serve different masters, so two profits arise; the gap is fully explainable. Three designs: **Non-integrated/interlocking** (two separate self-balancing ledgers + Cost Ledger Control A/c; needs reconciliation), **Integrated/integral** (one ledger, real financial accounts, NO CLC, NO reconciliation), and the **Reconciliation Statement** bridging cost vs financial profit. Degree of integration: full (residual reconciliation = 0) or partial (shared up to prime/works cost).

Core concepts

- **Cost Ledger Control A/c (CLC)** = General Ledger Adjustment A/c = proxy for the entire financial world; receives the "other leg" of every externally-originating transaction so the cost ledger self-balances.
- **Self-check:** CLC balance = net total of all other control-account balances (= net assets/stock inside cost ledger).
- Opening asset balances (Stores/WIP/FG): Dr in own account, Cr to CLC → CLC opens as credit balance.
- Interlocking = separate + interval reconciliation; Integral = one book. Memorandum reconciliation = a presentation format (T-account), not a system, applies only to non-integrated.
- Integrated: notional charges cannot enter a real-account ledger → dropped or kept as memorandum → "no reconciliation" and "no notional charges" go together.

Key provisions / rules

Non-integrated key entries (external leg → CLC):

Transaction	Entry
Materials purchased	Stores Ledger Control Dr / To CLC
Direct material issued	WIP Control Dr / To Stores
Indirect material	Production OH Control Dr / To Stores
Wages paid	Wages Control Dr / To CLC
Direct/Indirect wages	WIP / Production OH Dr / To Wages Control
Overheads incurred	OH Control Dr / To CLC
Production OH absorbed	WIP Control Dr / To Production OH Control
Goods completed	Finished Goods Control Dr / To WIP
Cost of goods sold	Cost of Sales Dr / To Finished Goods
Sales	CLC Dr / To Costing P&L
Over-absorption	Production OH Control Dr / To Costing P&L
Under-absorption	Costing P&L Dr / To Production OH Control
Costing profit	Costing P&L Dr / To CLC
Notional charge	OH Control Dr / To CLC (→ later a reconciling item)
Abnormal loss	Costing P&L Dr / To Stores/WIP

Integrated: internal cost legs identical; only the outside leg changes from CLC to real accounts (Sundry Creditors, Bank, Sundry Debtors, Provision for Depreciation, Outstanding Expenses). Under/over-absorption → single P&L. Proof = one balanced trial balance.

Over/Under absorption: Absorbed − Actual = **+over** (costing gain) / **−under** (costing loss). Applies to factory, admin AND selling OH.

Four families of reconciling items:

- A. Financial-only items (interest paid, loss on asset sale, goodwill/preliminary w/off, penalties, donations, tax provision, dividends paid; financial incomes: interest/dividend/rent received, profit on asset sale; appropriations: reserves/sinking fund).
- B. Cost-only notional charges (notional rent/interest on capital/proprietor's salary).
- C. Over/under-absorption of overheads (all three types).
- D. Different valuation/methods (stock FIFO vs WA; depreciation SLM vs machine-hour; WIP basis).

Reconciliation direction — start from COST profit, reach FINANCIAL:

ADD (financial profit higher)	SUBTRACT (financial profit lower)
Financial incomes (interest/dividend/rent recd, profit on asset sale)	Financial expenses (interest paid, loss on sale, fines, donations, goodwill/preliminary w/off, tax provision, dividend paid, reserves)
Overhead **over**-absorbed in cost	Overhead **under**-absorbed in cost
Notional charges in cost only	—
Cost depreciation **higher** than financial	Cost depreciation **lower** than financial
Closing stock **lower** in cost / Opening stock **higher** in cost	Closing stock **higher** in cost / Opening stock **lower** in cost

Mirror rule: start from FINANCIAL profit → flip every sign. Always label the anchor. **Stock direction:** ↑ closing stock ⇒ ↑ profit; ↑ opening stock ⇒ ↓ profit (opposite directions). **Two-step test:** (1) which book, raised or lowered its profit? (2) starting from the other book, move to reproduce that effect.

Worked mini-example

Costing profit ₹2,50,000. Works OH under-recovered 6,000 (−); Admin OH over-recovered 4,000 (+); interest received 8,000 (+); dividend received 5,000 (+); loss on sale of machinery 10,000 (−); provision for income tax 30,000 (−); notional rent charged in cost 12,000 (+); preliminary exp w/off 3,000 (−). Add = 4,000+8,000+5,000+12,000 = 29,000; Less = 6,000+10,000+30,000+3,000 = 49,000. **Financial Profit = 2,50,000 + 29,000 − 49,000 = ₹2,30,000.**

Exam traps & must-remember

- Wrong direction of adjustment (#1 error) — always re-anchor.
- Over vs under reversed: write "Absorbed − Actual" and read the sign. Over-absorbed = costing gain → subtract to reach financial.
- Stock: higher closing ⇒ higher profit; higher opening ⇒ lower profit (opposite). Restate each stock line as a profit effect.
- Notional items are cost-only → add back from cost profit.
- Integrated system has NO CLC and needs NO reconciliation; using CLC there is a red flag.
- Provision for tax, dividends, transfers to reserves = financial-only appropriations — easy to forget.
- "Equal in both books" ⇒ no reconciling entry.
- Reconcile admin & selling absorption too, not just factory.
- Ledger opening stocks route through CLC (asset Dr / CLC Cr); omit them and the ledger won't tie.
- Never force a balancing figure — the statement derives the second profit; if both given and they don't tie, items govern.
- Three interests: received (add), paid (subtract), notional on own capital (add back).

One-line recall

- Non-integrated = separate ledger + CLC + reconciliation; Integrated = one ledger, real accounts, no CLC, no reconciliation.
- CLC balance = sum of all other cost-ledger balances (built-in tie-out).
- Reconciling items = 4 families: financial-only, cost-only notional, over/under-absorption, valuation differences.

- From cost profit: add financial incomes/over-absorption/notional/higher-cost-charges; subtract financial expenses/under-absorption.
- Stock: ↑closing ⇒ ↑profit, ↑opening ⇒ ↓profit.
- Mirror rule flips all signs when starting from financial profit.

Chapter 08 — Unit & Batch Costing

Snapshot

Two methods on the homogeneity spectrum. **Unit/Output/Single costing** — one homogeneous product, mass/continuous production (bricks, cement, steel, sugar, paper, power): divide period cost by period output via a Cost Sheet. **Batch costing** — identical articles in distinct lots (pharma, bakery, garments, printing, components): unit costing inside the batch, job costing between batches; the batch is the cost unit. **EBQ** decides optimum batch size. Decision test: "did these units consume resources identically?" All period → unit; per lot → batch; each order bespoke → job.

Core concepts

- Division is honest only between identical things; grades/by-products need weighting, not blind division.
- "Cost per unit" is a family: prime/works/production/sales per unit — always ask "at which stage?"
- Setup cost is **fixed per batch/event** (same rupees whether 100 or 100,000 pieces) → makes batch size matter → drives EBQ. Remove setup or holding cost and EBQ collapses.
- EBQ = EOQ wearing a factory uniform (setup replaces ordering cost).

Key provisions / rules

Cost Sheet ladder: Prime → (+Factory OH ±WIP −Normal scrap) Works Cost → (+Admin OH) Cost of Production → (±FG stock) COGS → (+S&D OH) Cost of Sales → (+Profit) Sales.

Key formulas:

- Direct Material Consumed = Op. RM + Purchases + Carriage In − Returns − Cl. RM
- Prime Cost = Direct Material + Direct Labour + Direct (Chargeable) Expenses
- Works Cost = Gross Works Cost + Op. WIP − Cl. WIP
- COGS = Cost of Production + Op. FG − Cl. FG
- Units sold = Op. FG + Produced − Cl. FG
- Profit p% on sales: Sales = Cost of Sales ÷ (1 − p); p% on cost: Sales = Cost × (1 + p)

Stock stages: RM before Prime; WIP at Works (valued at factory cost); FG at Cost of Production → COGS (valued at COP/unit). Closing FG at current year's COP/unit; opening FG at last year's rate (unless FIFO/WA specified).

Scrap: normal scrap sale deducted from works/factory cost; abnormal loss → Costing P&L.

Carriage: inward = material cost (before prime); outward = S&D (after COGS).

Batch costing:

- Total Batch Cost = Direct Materials + Direct Labour + Setup cost + Overheads absorbed.
- **Cost per unit = Total Batch Cost ÷ number of GOOD units** (normal rejects: divide by good units; abnormal spoilage stripped out to Costing P&L at cost less scrap).
- OH absorbed by labour/machine-hour rate OR % of direct cost — apply each block to its own base.
- Batch may span several orders: compute cost per good unit, then × order units.

EBQ:

- **EBQ = √(2 D S ÷ C)** (D = annual demand, S = setup cost/batch, C = holding cost/unit/annum; if C = i% of price p, use C = i·p)
- **EBQ (gradual replenishment) = √(2DS ÷ [C(1 − d/p)])** — only when both production rate p and usage rate d given; gives a larger batch.
- Number of batches/year = D ÷ EBQ; Cycle = (EBQ ÷ D) × 360 days (or ×12 months, ×365).
- At EBQ: annual setup cost = annual holding cost (self-check).

- **Minimum total relevant cost = √(2 D S C)** = twice the setup (= twice the holding) at optimum.
- Curve is flat-bottomed → near-optimum round sizes cost almost nothing extra.
- Halving setup cost shrinks EBQ and total relevant cost by factor √2 (lean/SMED lever).

Excluded (financial) items: interest on loan/debentures, income tax, dividends, donations, transfer to reserves, loss/profit on sale of assets/investments, goodwill written off, fines, cash discount.

Worked mini-example

Annual demand 24,000 strips; setup ₹1,200/batch; holding ₹4/strip p.a.

- EBQ = √(2×24,000×1,200 ÷ 4) = √1,44,00,000 = **3,795 strips**.
- Batches/year = 24,000 ÷ 3,795 = 6.3; Cycle = (3,795 ÷ 24,000) × 360 = 57 days.
- Check: setup (24,000/3,795)×1,200 = ₹7,589 ≈ holding (3,795/2)×4 = ₹7,590 ✓; total = √(2×24,000×1,200×4) = ₹15,179.
- Q=4,000 → total ₹15,200 (worse); Q=3,000 → ₹15,600 (worse).

Exam traps & must-remember

- Purchases ≠ Material Consumed (adjust RM stock + carriage inward).
- WIP adjusted at Works, FG at COGS — never swapped.
- Closing FG at current year's COP/unit, not selling price/last year's rate.
- Cost of production ÷ units produced; cost of sales/profit relate to units sold.
- 20% on sales ⇒ cost = 80% of sales; 20% on cost ⇒ Sales = Cost × 1.20 (note: 20% on sales = 25% on cost).
- Divide batch by GOOD units (not units started) for normal rejects; abnormal loss to P&L (value at total-input rate less scrap).
- EBQ: convert holding % to ₹ (C = i·p); the Q/2 halving is already in the formula — don't halve again.
- Don't eyeball square roots — square the answer back (√1,44,00,000 = 3,795, not 4,000).
- Gradual-replenishment (1 − d/p) form ONLY when both production and usage rates given.
- "Cost of one batch" includes that batch's setup; annual trade-off uses (D/Q)·S — same S, two roles.
- Read the base of a % overhead (of wages vs prime cost vs works cost).
- Opening FG at last year's rate; only closing FG at current rate.

One-line recall

- Unit costing = period cost ÷ period output (homogeneous mass output).
- Batch cost/unit = total batch cost ÷ good units; batch = the "job".
- EBQ = √(2DS/C); at optimum setup cost = holding cost; min total = √(2DSC).
- EBQ is EOQ with setup replacing ordering cost; flat-bottomed curve.
- Stock stages: RM before Prime, WIP at Works, FG at COP→COGS.
- Halving setup ⇒ EBQ and total cost fall by √2.

Chapter 09 — Job & Contract Costing

Snapshot

When output is heterogeneous, averaging lies — so tag every cost to a unique identifier and never average across identifiers. **Job costing** — custom, one-off orders (printers, garages, fabricators); each order a distinct cost object. **Batch costing** — identical units in lots (batch = the job); cost/unit = batch cost ÷ units; EBQ optimises lot size. **Contract costing** — large, long, site-based jobs crossing accounting years, kept in the contractor's books, dominated by direct cost; needs prudence machinery for interim profit.

Core concepts

- Job/Batch/Contract = one family, three sizes; decide by "which identifier owns this rupee?"
- Contract needs extra machinery because contracts cross years, are mostly direct cost, and payment is controlled by client certification.
- Prudence: **anticipate no profit, provide for all losses.** Recognise only a fraction of notional profit; the fraction grows with completion and is discounted again for cash received (retention risk).
- Job costing WIP = total of open (unfinished) job cost sheets.

Key provisions / rules

Job Cost Sheet: Prime (DM+DL+Direct Exp) → +Works OH = Works Cost → +Admin OH = Cost of Production → +S&D OH = Total Cost → +Profit = Price.

- Profit on **cost** m%: Price = Cost × (1+m). Profit on **sales** m%: Price = Cost ÷ (1−m).
- Ledger: Dr WIP Control / Cr Stores, Wages, Production OH Control; on completion Dr Finished Goods (or Cost of Sales) / Cr WIP. Job carries absorbed (predetermined) OH.
- Spoilage from general cause → overhead (all jobs); from customer's special specification → charged to that job.

Batch/EBQ: EBQ = √(2DS ÷ C); setups/yr = D ÷ EBQ; at optimum total setup cost = total carrying cost; convert C to ₹ if given as % of cost; (1−d/p) factor only if both rates given.

Contract vocabulary:

- **Contract Price** = finish line, not revenue-to-date.
- **Work Certified** = value (at contract/selling price, incl. profit) of work approved by architect/surveyor to date; anchor for everything.
- **Work Uncertified** = done but not yet certified; carried at COST (no profit).
- **Retention Money:** Cash Received = Work Certified × (1 − retention%); Retention = Certified − Cash.
- **Notional Profit = Value of Work Done − Cost of Work Done**, where Value = Work Certified + Work Uncertified, and Cost of Work Done = costs incurred to date − closing (materials at site + plant WDV).
- **Estimated Total Profit = Contract Price − (Cost to date + Estimated cost to complete).**
- **% Completion = Work Certified ÷ Contract Price × 100.**

Contract Account: Dr = materials, wages (incl. accrued), direct expenses, plant (cost or depreciation), sub-contract, extra work, apportioned HO OH. Cr = materials at site c/d, plant WDV c/d, materials sold, WIP (Work Certified + Work Uncertified). Balancing figure = Notional Profit/Loss.

Six fates of material: consumed / at-site c/d / returned to stores / transferred to another contract / sold (profit-loss on sale → P&L) / lost (abnormal → P&L).

Plant: Method 1 (short) — debit plant at cost, credit closing WDV (difference = depreciation charged). Method 2 (long) — debit only depreciation, no cost-in/WDV-out. Hired plant — only hire charges debited. Both methods give identical notional profit.

Prudence ladder (profit to P&L):

Completion	Transfer to P&L
< 25%	NIL
25% to < 50%	⅓ × NP × (Cash ÷ Certified)
50% to < 90%	⅔ × NP × (Cash ÷ Certified)
≥ 90%	Est. Profit × (Cash ÷ Contract Price) [= Est.P × (Certified/Price) × (Cash/Certified)]; or cost-ratio variant Est.P × (Cost to date ÷ Est. total cost) [× Cash/Certified]
Any % with **notional loss**	**Full loss immediately** (no fraction, no cash ratio)
Any % with **overall estimated loss** (Est. total cost > Contract price)	**Full estimated loss immediately**

Bands are conventions — if question states a policy, follow it; else state your assumption.

Balance Sheet WIP = (Certified + Uncertified) − Reserve − Cash received = Cost of work done + Profit recognised − Cash received. Reserve = Notional Profit − Profit recognised. Also show materials at site, plant WDV as assets; accrued wages as liability.

Escalation clause (protects contractor; de-escalation protects client):

- **Claim = Σ [Standard Qty allowed × (Actual Rate − Base Rate)]** — rate rise on STANDARD quantity only, never on wastage; net de-escalation if clause symmetric. Standard qty is for work actually done. Added to contract price.
- **Cost-plus contract:** price = allowable cost + agreed profit (fixed fee or % of cost); shifts inflation risk to client; cost-plus-% rewards overspending so clients cap/prefer fixed-fee; contractor's books open to client audit.

Worked mini-example

Contract price ₹40,00,000. Cost of work done ₹20,50,000; Value of work done (Certified 24,00,000 + Uncertified 60,000) = 24,60,000 → Notional Profit ₹4,10,000. Cash received 85% = ₹20,40,000.

- % completion = 24,00,000 ÷ 40,00,000 = 60% → ⅔ band.
- Profit to P&L = ⅔ × 4,10,000 × (20,40,000 ÷ 24,00,000) = ⅔ × 4,10,000 × 0.85 = **₹2,32,333**.
- Reserve = 4,10,000 − 2,32,333 = ₹1,77,667.
- BS WIP = 24,60,000 − 1,77,667 − 20,40,000 = ₹2,42,333 (= 20,50,000 + 2,32,333 − 20,40,000 ✓).

Exam traps & must-remember

- Compute % completion FIRST; don't grab ⅔ reflexively (20%→NIL, 40%→⅓, 92%→estimated basis).
- **Check overall loss BEFORE choosing a band** — if Est. total cost > Contract price, provide full estimated loss now regardless of % (highest-value trap).
- Never drop × (Cash ÷ Certified) in ⅓/⅔ formulas — plant a non-standard retention % to catch this.
- Add accrued/outstanding wages to debit.
- Plant: one method only (cost-in requires WDV-out; depreciation basis needs neither). Mid-year arrival → depreciate for months actually at site.
- Work Uncertified at cost (no profit); only Work Certified carries selling-price value.
- Loss transferred in full immediately — no ⅓/⅔, no cash ratio (asymmetry is the point).
- Escalation on standard qty × rate rise, never actual/wasteful qty; net de-escalation if symmetric.
- "Profit on cost" ≠ "profit on sales."
- Notional Profit (work to date) vs Estimated Profit (whole contract) — label separately.
- BS WIP: deduct BOTH reserve and cash received.
- EBQ: convert carrying-% to ₹.
- Lump-sum advance ≠ cash on certification (treat advance as liability unless equated).

One-line recall

- Job = wallet per order; never average across identifiers.
- Notional Profit = Value of Work Done − Cost of Work Done; % completion = Certified ÷ Contract Price.
- Prudence ladder: <25% NIL, 25–50% ⅓, 50–90% ⅔, ≥90% estimated basis; all × (Cash/Certified).
- Losses full and immediate; overall estimated loss → provide entire loss now.
- Cash Received = Certified × (1 − retention%); BS WIP = (Certified+Uncertified) − Reserve − Cash.
- Escalation = Σ[Standard Qty × (Actual − Base Rate)]; EBQ = √(2DS/C).

Chapter 10 — Process & Operation Costing

Snapshot

For homogeneous, continuous, untraceable output: cost the process, not the unit, and divide at the end. **Cost per unit = Total cost poured in ÷ effective good units.** Everything else (normal/abnormal loss, equivalent units, FIFO/WA) exists to keep numerator and denominator on the same fair basis. Legal fit when output is homogeneous + production continuous/mass + cost cannot be economically traced (chemicals, paint, cement, sugar, textiles, paper, oil refining, food, breweries, plastics, steel rolling). Standardised components in a customised product → **hybrid**.

Core concepts

- Each process = a T-account (cost centre); output of one becomes input of next (transfer at cost).
- Normal loss = planned/unavoidable → absorbed by good units, valued at scrap only. Abnormal loss = failure signal → valued at full normal cost, exposed in Costing P&L.
- Abnormal gain = actual loss < normal → debited back to process at normal cost; must return the scrap it never earned.
- Equivalent units convert half-done work; completion measured **per cost element** (material usually 100%, conversion partial).
- One equation, four disguises — same division, different denominator.

Key provisions / rules

Cost per good unit (no WIP) = (Total process cost − Scrap value of normal loss) ÷ (Input units − Normal loss units).

Normal loss: units = % × input (read the base — "of input" vs "of output" vs "inspected"; solve algebra if of output). Valued at scrap value only (nil if not saleable); **zero EU**; scrap deducted from material cost before dividing.

Abnormal loss = Expected good output − Actual output (= Actual loss − Normal loss). Valued at **normal cost/unit**. Credit process → Abnormal Loss A/c; abnormal units still fetch scrap (credited inside Abnormal Loss A/c, not Normal Loss A/c); net → Costing P&L.

Abnormal gain = Actual output − Expected good output. Valued at **normal cost/unit**; **debit** process. Abnormal Gain A/c debited with scrap value NOT realised (to Normal Loss A/c), net gain → Costing P&L. Always at normal rate, never a fresh period rate.

Equivalent units = Physical units × % completion, per element. WIP defaults: material added at start → 100% for material; conversion accrues evenly → to stated stage; a second material added at 75% stage → 0% for WIP below that stage.

Weighted Average vs FIFO:

	Denominator (EU)	Cost used
Weighted Avg	Completed 100% + Closing WIP EU	Opening WIP cost + current cost
FIFO	(Opening WIP × % remaining to complete) + Started-&-completed + Closing WIP EU	Current period cost only

- FIFO completed-goods cost = Opening WIP cost b/f + cost to complete it + (Started & completed × current rate).
- FIFO three layers: finish opening + do middle whole + start closing.
- Rising prices: FIFO rate > WA rate → FIFO transfers out slightly less, leaves dearer closing stock; stable prices → converge.

WIP + loss together: normal loss = physical units in reconciliation but **zero EU**, scrap deducted from material element; abnormal loss = EU line at its degree of completion (100% if detected at end; part-way if inspection earlier); abnormal gain = EU line (usually 100%) subtracted and debited back.

Statement order: Equivalent Production → Cost per EU → Evaluation → Process A/c. **Process A/c layout:** Dr = Materials, Wages, Expenses, Overhead, **Abnormal Gain**. Cr = **Normal Loss (scrap)**, **Abnormal Loss**, **Transfer/Finished**, **Closing WIP**. Units columns must tie before rupees.

Inter-process profit: keep **Cost / Profit / Total** columns.

- Unrealised profit in closing stock = Closing stock value × (Profit in total column ÷ Total column value).
- Balance-sheet stock = Total − Unrealised profit; create **Stock Reserve** for the difference.
- Realised profit = Sum of process profits − Unrealised profit in closing stocks (nets to zero on consolidation — pure management-info device).
- Watch "profit on cost" (×margin) vs "profit on transfer/selling price" (÷[1−margin]).

Operation costing (finer resolution): cost per operation = (operation cost − scrap) ÷ good units of that operation; **finished-unit cost = Σ operation costs**. Same loss/EU logic per operation; excellent per-step control benchmark.

Waste (no value, shrinks denominator) vs **scrap** (small recoverable value, reduces numerator) vs **spoilage** (unrectifiable, sold as seconds) vs **defectives** (rectifiable — normal rectification = production cost, abnormal → P&L).

Worked mini-example

Input 2,000 @ ₹5 = 10,000; wages 5,000; OH 4,000. Normal loss 10% of input, scrap ₹5/unit. Actual output 1,700.

- Normal loss = 200 units (scrap ₹1,000); expected good = 1,800.
- Cost/unit = (19,000 − 1,000) ÷ (2,000 − 200) = 18,000 ÷ 1,800 = **₹10**.
- Abnormal loss = 1,800 − 1,700 = 100 units × ₹10 = ₹1,000.
- Abnormal Loss A/c: To Process 1,000 / By Bank (scrap 100×5) 500, By Costing P&L 500.

Exam traps & must-remember

- Never value normal loss at full cost — scrap value only.
- Denominator = (input − normal loss), not input; deduct scrap from numerator first.
- Abnormal loss AND gain valued at normal cost/unit, never at scrap.
- Abnormal-gain scrap correction: Normal Loss and Abnormal Gain accounts exchange scrap of units NOT actually lost.
- Abnormal gain = debit to process; abnormal loss = credit. Reversing = red flag.
- FIFO: opening WIP contributes only % remaining, uses current-period cost only; opening cost added as a lump when valuing completed goods.
- Completion measured per element (material 100%, conversion partial).
- With WIP + abnormal items, abnormal units appear in EU statement at completion; normal loss = zero EU.
- Inter-process unrealised profit uses profit-to-total ratio of goods available, not the transfer margin; show stock net of it.
- Abnormal-loss scrap recovery credited inside Abnormal Loss A/c, not process.
- "10% of input" ≠ "10% of output" ≠ "10% inspected" — solve algebra when base is output.
- "Profit on cost" vs "on transfer/selling price" decides the divisor.
- Normal-loss scrap reduces the MATERIAL element cost, not conversion.
- Carry per-EU rate unrounded; round only final figures.
- Transfer-in cost = its own debit line ("To Process I A/c"), 100% complete for receiving process; not merged into materials added.

One-line recall

- Cost/unit = (Total cost − normal-loss scrap) ÷ (Input − normal-loss units).
- Normal loss: scrap value only, zero EU; abnormal loss/gain: normal cost/unit, to P&L.
- Abnormal loss = credit process; abnormal gain = debit process (returns unearned scrap).
- WA pools opening+current cost; FIFO uses current cost + only work done this period.

- EU = physical units × % completion, per element; statement order EP→Cost/EU→Evaluation→Process A/c.
- Inter-process unrealised profit = stock × (profit ÷ total of goods available); operation cost = Σ per-operation costs.

Chapter 11 — Joint Products & By-Products

Snapshot

Where one process yields several products simultaneously, the pre-split-off cost is **joint** and cannot be traced — it must be **apportioned by a convention**. Post-split-off cost is **separable**, traced directly. By-products (minor value) get **no** joint cost pushed onto them; their net realisation is **credited back** to joint cost. Apportioned joint cost is **sunk** and irrelevant to further-processing decisions.

Core concepts

- **Split-off point**: point where products become separately identifiable.
- **Joint cost**: total cost up to split-off (only ever *apportioned*, never *allocated*).
- **Separable / subsequent cost**: post-split-off cost, *allocated* (traced) to one product.
- **Joint product**: significant value, produced simultaneously in fixed/uncontrollable ratio. **By-product**: minor, incidental. **Co-product**: made together but controllable mix / not necessarily same raw material.
- Value ranking & treatment table:

Term	Value	Treatment
Joint/main product	High	Bears apportioned joint cost
By-product	Minor but saleable	Net realisation **credited to** joint cost; none pushed onto it
Scrap	Small residual material	Net realisation credited to job/process
Waste	Nil/negative	Disposal cost **added** to process cost

Key provisions / rules

Apportionment methods

Method	Basis	Best when
Physical units (average)	Quantity (common unit)	Products similar value/unit
Weighted / survey / points	Physical units × point factor	No prices but richness differs
Market value at split-off	Qty × split-off price	Split-off price exists
NRV (workhorse)	Final sales − separable & selling	Products processed further
Constant gross-margin	Force equal GP%	Products deemed equally profitable

Formulas:

- Physical units: Cost per unit = Total joint cost ÷ Total units; Product share = rate × its units. Apportion on **output at split-off**, not input; normal-loss units get no cost.
- Weighted: Weighted units = physical output × point factor; Product share = Total joint cost × (its weighted units ÷ total weighted units). Rate is **per weighted unit**.
- Market value at split-off: ratio = qty × split-off price (use total value, not price alone).
- NRV at split-off = Final sales value − separable (post-split) costs − selling/distribution costs. Then apportion in NRV ratio. **Do NOT deduct notional profit** in joint-product NRV.
- Constant gross-margin steps: (1) Overall GP% = (Total sales − Total costs[joint+all separable]) ÷ Total sales; (2) GP each = its sales × GP%; (3) Total cost each = sales − GP; (4) **Joint cost each = total cost − its separable cost**. Can give tiny/negative allocation — do not floor at zero.

By-product treatment

- (A) Non-cost: (i) Other income — net sales to Costing P&L (main product cost per unit *unaffected*); (ii) **NRV credited to process** (net realisation deducted from joint cost) = examiner-preferred default; lowers main product cost per unit.

- (B) Cost methods: opportunity/replacement cost (if consumed internally); standard cost; **reverse-cost/working-back** (from sale value, subtract profit + selling + processing to get notional joint-cost credit).
- Always credit **net** (sales − by-product's own selling AND processing costs), never gross.

Further-processing decision (relevant costs only) Process further only if: (Sales value after − Sales value at split-off) > Further-processing cost.

- Apportioned joint cost is **SUNK — ignore**.
- **Avoidable separable fixed cost** caused by processing **IS** relevant.
- Incremental revenue = qty × (new price − split-off price).

Worked mini-example

Joint cost ₹2,00,000; P 5,000 L, Q 4,000 L; by-product R 1,000 L sells ₹8, selling cost ₹1. R net realisation = 1,000 × (8−1) = ₹7,000 → net joint cost = 2,00,000 − 7,000 = ₹1,93,000. Apportion by split-off value: P 5,000×40 = 2,00,000; Q 4,000×30 = 1,20,000; ratio 0.625:0.375. P = 1,93,000×0.625 = ₹1,20,625; Q = ₹72,375. (Sum = 1,93,000 ✓) Further processing: P incremental rev 5,000×(50−40)=50,000 vs cost 30,000 → +20,000 process; Q 4,000×(35−30)=20,000 vs 25,000 → −5,000 sell at split-off.

Exam traps & must-remember

1. Use **NRV not final sales** when product processed after split-off.
2. **Never** bring joint cost into further-processing decision (sunk); it's a distractor.
3. **Never** apportion joint cost onto by-product; credit its net realisation back.
4. Credit **net** by-product realisation (minus selling AND processing costs).
5. Mixed units → convert to common measure first.
6. Negative joint-cost under constant-margin is correct — state, don't adjust.
7. **Reconcile**: Σ apportioned = given joint cost; Σ product profits = overall profit.
8. Value decides classification, not the label.
9. After constant-margin, all products show equal margin — don't name a "winner".
10. Market value **at split-off** ≠ final price; use split-off value even if later processed.
11. Reverse-cost: "25% on sales" ≠ "25% on cost" (= 33⅓% on cost). Convert base.
12. Use **output at split-off**, not input quantity, as denominator.
13. After weighted units, rate is per weighted unit — don't re-multiply by physical output.

One-line recall

- Left of split-off = joint (apportion by convention); right = separable (trace directly).
- NRV = Final sales − separable − selling; apportion in NRV ratio (no profit deducted).
- By-product: credit **net** realisation to joint cost; push no cost onto it.
- Process further iff incremental revenue > further-processing cost; joint cost is sunk.
- Constant gross-margin: joint cost each = (sales − GP@overall%) − separable; negatives allowed.
- Always reconcile apportioned joint cost to the total given.

Chapter 12 — Service Costing

Snapshot

Service (operating) costing measures the cost of an **intangible, non-storable, instantly-consumed** output. No product → no stock valuation, no WIP. The whole skill is **choosing the right denominator** — usually a **composite cost unit** fusing a quantity dimension × a service dimension (tonne-km, passenger-km, patient-day, kWh, room-day). Cost is **capacity/standing-charge dominated**, so occupancy/utilisation drives cost per unit.

Core concepts

- **Composite (compound) cost unit** = quantity dimension × service dimension. Use when both dimensions vary and matter; else a **simple unit** (per meal, per ticket, per kilolitre) suffices.
- **Internal (captive) service** (boiler, captive power, transport pool, canteen): recover **cost, not price** — deliverable is a **transfer/recovery rate**; usually **no profit margin**. **External** service: cost feeds a fare/tariff with margin.
- Behaviour identity: Cost per unit = (F + vN) ÷ N = F/N + v. Fixed term collapses as N (utilisation) rises → "occupancy is destiny".

Key provisions / rules

Core formula: Cost per composite unit = Total operating cost ÷ Total composite units.

Cost classification (transport template)

- **Standing (fixed)**: time-based depreciation, insurance, road tax, permit, garage/depot rent, monthly salaries, admin/supervision, licence fees, interest on capital.
- **Running (variable)**: fuel, lubricating oil, tyres/tubes, mileage-based depreciation, per-trip/per-km/commission wages.
- **Maintenance (semi-variable)**: repairs, spares, servicing, painting, overhauls.

Tonne-km / passenger-km — two builds

- **Absolute (weighted)** = Σ(load on each leg × distance of that leg). **Default for cost.**
- **Commercial (average)** = Average load × Total distance. Use only if stated.
- Heavy loads on long legs ⇒ absolute > commercial; on short legs ⇒ absolute < commercial; equal when load same each leg.

Denominator recipe (transport)

1. Effective days = total days − idle/maintenance days.
2. Distance = days × trips × km per trip (×2 for return legs of a round trip).
3. Capacity used = seats × load factor (or tonnes × utilisation).
4. Composite units = Σ(load × distance) per leg (absolute).

Offered vs sold: Cost per km/seat-km (offered) = efficiency; Cost per passenger-km (sold) = pricing = cost per seat-km ÷ load factor.

Pricing (watch the base)

- Profit on **cost**: Fare = Cost × (1 + m).
- Profit on **takings/sales**: Fare = Cost ÷ (1 − m).

Other sectors

- Hospital: patient-days = beds × days × occupancy% (+ hired beds × days occupied); include hire charge in cost.
- Hotel: occupied room-days = rooms × days × occupancy% (per season); divide by **occupied**, not available.

- Power/steam/water: divide cost by units **delivered/consumed** (net of transmission/evaporation loss), not units generated.

Depreciation flag: per-annum/straight-line ⇒ **standing**; per-km/mileage ⇒ **running**.

Worked mini-example

Bus: 50 km one-way, 40 seats, 2 round trips/day, 25 days, 80% load factor. Distance = 50×2×2×25 = 5,000 km. Passengers = 40×80% = 32. Passenger-km = 5,000×32 = 1,60,000. Total operating cost = ₹1,18,000. Cost/passenger-km = 1,18,000 ÷ 1,60,000 = ₹0.7375. Fare at 20% on takings = 0.7375 ÷ 0.80 = ₹0.9219/passenger-km. Check: takings 0.9219×1,60,000 = ₹1,47,500; profit 29,500 = 20% of takings ✓. (If "20% on cost": fare = 0.7375×1.20 = ₹0.885 — different answer.)

Exam traps & must-remember

1. **Depreciation basis** — time ⇒ standing; km ⇒ running. #1 trap.
2. **Profit on cost vs on takings**: ÷(1−m) for takings, ×(1+m) for cost.
3. Count **both legs** for distance/fuel, but tonne-km only for **loaded** leg (+ back-load).
4. **Absolute vs commercial** — default absolute unless "commercial" stated.
5. Use **occupied** capacity, not available, when occupancy given.
6. Subtract **idle/maintenance days** to get effective running days.
7. "2 round trips" = 4 one-way legs.
8. Hired beds/rooms add to **both** cost and denominator.
9. Wage basis decides bucket: monthly salary = standing; per-trip/km = running.
10. Convert fuel: ₹90/L at 4 km/L = ₹22.50/km (two-step).
11. Simple vs composite: don't cost "per patient" or "per tonne" when a dimension varies.
12. Power/water/steam: divide by units **delivered**, not generated (loss recovered from what arrives).
13. **Include interest on capital** if given (standing charge); don't drop as "financial".
14. **No margin** on internal/captive service unless output sold externally.
15. Never average per-unit rates across mixed fleet/bed-class/season — build total cost & total units, divide once (or weight by volumes).

One-line recall

- Cost per composite unit = Total operating cost ÷ Total composite units.
- Composite unit = quantity dimension × service dimension.
- Absolute tonne-km = Σ(load × distance per leg); default over commercial.
- Fare = Cost ÷ (1 − m) on takings; = Cost × (1 + m) on cost.
- Divide by units delivered/occupied, not generated/available.
- Depreciation: per-annum = standing, per-km = running.

Chapter 13 — Standard Costing & Variance Analysis

Snapshot

Set a pre-determined "should be" cost per unit, compare with actual, and **split the gap into named pieces** each traceable to one cause/owner. Universal engine: **change one thing at a time** — price variance holds quantity at **actual**, usage variance holds price at **standard**. Every parent = sum of its children (the reconciliation is the proof). SQ/SH are always for **actual output**.

Core concepts

- **Standard hour** = quantity of output that should be produced in one hour (output expressed in time).
- **Three-column skeleton** (material/labour/VOH): ① SQ×SP · ② AQ×SP · ③ AQ×AP → Usage/Efficiency = ①−②, Price/Rate = ②−③, Cost = ①−③.
- **Sign convention**: Cost variances = **Standard − Actual** (positive = Favourable). Sales = **Actual − Budget**. Master rule: **Favourable = increases profit**.
- **Joint variance** (SP−AP)×(SQ−AQ) is absorbed into the **price** variance (why price uses actual quantity).
- Standard types: **Ideal** (perfect, demotivating), **Basic** (fixed long-term, goes stale), **Current**, **Normal/Attainable** (preferred). Revise standards only for **permanent** changes.
- Budgetary control = totals/any function (extensive); standard costing = per-unit, needs measurable repetitive output (intensive/deeper).

Key provisions / rules

Let SQ/SH = standard qty/hours **for actual output**; AQ/AH = actual; SP/SR = standard price/rate; AP/AR = actual; RSQ/RSH = revised standard (actual total in standard ratio); AI = qty purchased.

Direct Material

- MCV = (SQ×SP) − (AQ×AP)
- MPV = (SP − AP) × AQ [use AI if price isolated at purchase point]
- MUV = (SQ − AQ) × SP → **MPV + MUV = MCV**
- MMV = (RSQ − AQ) × SP
- MYV = (SQ − RSQ) × SP → **MMV + MYV = MUV**
- RSQ = actual total input × standard mix ratio.
- Yield shortcut: MYV = (Standard output from actual input − Actual output) × Std cost per unit of output.
- Purchase-point: MPV on AI, stock at standard; MPV+MUV ≠ MCV (gap sits in closing stock).

Direct Labour (AH_paid; AH_worked = paid − idle)

- LCV = (SH×SR) − (AH_paid×AR)
- LRV = (SR − AR) × AH_paid [hours **paid**]
- LEV = (SH − AH_worked) × SR [hours **worked**]
- ITV = Idle hours × SR = (AH_paid − AH_worked) × SR — **always Adverse**
- → **LRV + LEV + ITV = LCV** (ITV=0 if no idle)
- LMV = (RSH − AH_worked) × SR; LYV = (SH − RSH) × SR → **LMV + LYV = LEV**
- Mix/yield use hours **worked** (strip idle first).

Variable Overhead (SR_v = std rate/hour)

- VOH Cost = (SH×SR_v) − Actual VOH
- VOH Expenditure = (AH_worked×SR_v) − Actual VOH
- VOH Efficiency = (SH − AH_worked) × SR_v → **Exp + Eff = Cost**
- Use hours worked (no VOH during idle).

Fixed Overhead (FOR = Budgeted FOH ÷ Budgeted output or hours; Absorbed = actual output × FOR)

- FOH Cost = Absorbed − Actual
- FOH Expenditure (Budget) = Budgeted − Actual
- FOH Volume = Absorbed − Budgeted → **Exp + Vol = Cost**
- FOH Capacity = (AH_worked − Budgeted hrs) × FOR/hr
- FOH Efficiency = (SH − AH_worked) × FOR/hr
- FOH Calendar = (Actual − Budgeted days) × Budgeted FOH/day → **Cap + Eff + Cal = Vol**
- Revised Budgeted Hours = Budgeted hrs/day × actual days (measure capacity against revised when calendar applies, to avoid double count).

Control ratios: Capacity = AH_worked ÷ Budgeted hrs; Efficiency = SH ÷ AH_worked; Activity = SH ÷ Budgeted hrs = Capacity × Efficiency. Above 100% = Favourable.

Sales — Margin (profit) method (reconciles profit; SM = std margin/unit, AM = actual margin/unit; RBQ = actual total sales × budget mix ratio)

- Total Sales Margin = (AQ×AM) − (BQ×SM)
- Price = (AM − SM) × AQ = (AP − BP) × AQ
- Volume = (AQ − BQ) × SM → **Price + Vol = Total**
- Mix = (AQ − RBQ) × SM; Quantity = (RBQ − BQ) × SM → **Mix + Qty = Vol**
- Marginal costing: SM = contribution/unit; **no FOH volume variance**.

Sales — Value method (does NOT reconcile to profit)

- Value = (AQ×AP) − (BQ×BP); Price = (AP − BP) × AQ; Volume = (AQ − BQ) × BP.

Reconciliation: Budgeted Profit ± Sales margin variances (→ standard profit on actual sales) ± Material ± Labour ± VOH ± FOH = **Actual Profit** (must tie exactly). Marginal: run budgeted→actual contribution, then subtract only FOH expenditure variance.

Worked mini-example

Std/unit: material 5 kg @ ₹40; labour 3 hr @ ₹50. Actual: 1,100 units; 5,720 kg cost ₹2,37,380; 3,410 hrs cost ₹1,74,910 (no idle). SQ = 1,100×5 = 5,500 kg; std mat = 2,20,000. SH = 3,300 hr; std lab = 1,65,000. MCV = 2,20,000 − 2,37,380 = 17,380 A. MPV = (40−41.50)×5,720 = 8,580 A. MUV = (5,500−5,720)×40 = 8,800 A. (Sum = 17,380 A ✓) LCV = 1,65,000 − 1,74,910 = 9,910 A. LRV = 1,74,910 − 3,410×50 = 4,410 A. LEV = (3,300−3,410)×50 = 5,500 A. (Sum = 9,910 A ✓)

Exam traps & must-remember

1. **Flex to actual output** for SQ/SH — never budgeted output. Biggest fatal error.
2. Rate/idle use **hours paid**; efficiency uses **hours worked**.
3. Idle time **always Adverse**.
4. Price on purchases (AI) vs usage (AQ) — read wording.
5. FOH Volume is **under-recovery not overspending**.
6. Mix uses (RSQ−AQ), Yield uses (SQ−RSQ); RSQ total must = AQ total.
7. Sales sign flips: favourable = actual exceeds budget.
8. Only **margin** method reconciles to profit (value method won't tie).
9. Compute rate/price as (Actual cost − AQ×SP) to avoid rounding AP/AR.
10. Calendar variance only when actual days ≠ budgeted; then capacity uses revised budgeted hours.
11. If reconciliation won't tie, a sign is flipped — diagnose.
12. FOR on **budgeted** activity, never actual/standard-for-actual.
13. Don't mix marginal (contribution volume) with absorption (FOH volume) in one answer.
14. SQ = standard input for output achieved; RSQ = standard-ratio split of actual input.

15. Efficiency valued at correct rate: labour SR, VOH SR_v, FOH FOR — same hour deviation.
16. Don't blame an **uncontrollable** variance on a manager.

One-line recall

- Price/rate holds quantity at actual; usage/efficiency holds price at standard.
- Cost variance = Standard − Actual; positive = Favourable; sales flips to Actual − Budget.
- Parent = sum of children at every node (Price+Usage=Cost, Mix+Yield=Usage, Cap+Eff+Cal=Vol).
- Idle time always adverse; rate uses paid hours, efficiency uses worked hours.
- FOR = Budgeted FOH ÷ Budgeted output; Absorbed = actual output × FOR; Volume = Absorbed − Budgeted.
- Activity ratio = Capacity × Efficiency; reconciliation lands exactly on actual profit.

Chapter 14 — Marginal Costing & CVP Analysis

Snapshot

For **short-run** decisions, total (full) cost lies — it fuses costs that move with activity and costs that don't. Marginal costing separates **variable** (relevant) from **fixed** (committed/sunk) and judges every choice on **contribution** = what actually changes. Master identity: **Contribution = Fixed cost + Profit**. Marginal values stock at **variable cost only** (fixed OH = period cost); absorption carries fixed OH into stock — the sole reason their profits differ when production ≠ sales.

Core concepts

- **Marginal cost** = aggregate variable cost = extra cost of one more unit (assumed constant per unit within **relevant range**).
- **Contribution/unit (c)** = Selling price − Variable cost/unit.
- Cost behaviour boxes: variable (per-unit constant), fixed (per-unit falls as volume rises), semi-variable (split by high-low). Fixed sub-types: **committed** (always ignore short-run), **discretionary** (avoidable), **stepped** (relevant if step crossed).
- Marginal vs absorption: only difference is whether **fixed factory OH sits in closing stock**. Over the business's life, both give **same total profit** (timing only). AS 2/Ind AS 2 require absorption (matching, normal capacity) for reporting; marginal is internal/decisions only.
- Marginal costing removes incentive to overproduce (profit moves with **sales**, not production).

Key provisions / rules

Contribution family

- Total contribution = Sales − Total variable cost = c × units = Fixed cost + Profit.
- Profit = Contribution − Fixed cost.

P/V ratio

- P/V = (Contribution ÷ Sales) × 100 = (c ÷ SP) × 100.
- P/V (two periods) = (Change in Contribution ÷ Change in Sales) × 100 = (Change in Profit ÷ Change in Sales) × 100.
- Variable-cost ratio = (VC ÷ Sales) × 100 = 100% − P/V.
- Improve P/V: raise price, cut variable cost/unit, shift mix to higher-P/V products. **Fixed cost does NOT affect P/V.**

Break-even & margin of safety

- BEP (units) = Fixed cost ÷ c.
- BEP (₹) = Fixed cost ÷ P/V ratio (= BEP units × SP).
- Cash BEP (units) = (Fixed cost − Non-cash fixed cost) ÷ c.
- MoS = Actual sales − BEP sales; MoS ratio = (MoS ÷ Actual sales) × 100 = Profit ÷ Contribution.
- Profit = MoS × P/V ratio; MoS (₹) = Profit ÷ P/V ratio.
- Profit as % of sales = P/V ratio × MoS ratio.
- BEP rises with fixed cost; falls with higher c; mix to higher-P/V lowers it; volume alone does NOT move BEP.

Target profit

- Units = (Fixed + Target profit) ÷ c; Sales = (Fixed + Target) ÷ P/V ratio.
- After-tax target: Pre-tax = After-tax ÷ (1 − tax rate).
- Profit = x% of sales: Sales = Fixed ÷ (P/V ratio − required % of sales).
- General: Profit = (Sales × P/V ratio) − Fixed cost.

High-low (split semi-variable)

- Variable/unit = (Cost at highest activity − Cost at lowest) ÷ (Highest units − Lowest units). Pick by **activity level**, not cost.
- Fixed = Total cost at a level − (Variable/unit × units at that level).

Multi-product (composite) BEP

- Composite P/V = (Total contribution ÷ Total sales) × 100; Composite BEP (₹) = Total fixed ÷ Composite P/V. Valid only for that exact mix.

Marginal vs absorption profit gap

- Gap = Fixed OH per unit × change in stock units (production − sales).
- Production > Sales → Absorption > Marginal; Production < Sales → Absorption < Marginal; equal when Production = Sales.
- Absorption also carries an **under/over-absorption** adjustment when production ≠ normal capacity (separate from stock-movement effect).

Decision rules (chase highest total contribution; check capacity first)

- Make-or-buy: **make if buy price > [marginal cost + avoidable fixed cost + opportunity cost of capacity]**.
- Special order: accept if price > variable cost/unit, given spare capacity & separate market; if full, add opportunity cost of displaced sales; order-specific fixed cost is relevant.
- Limiting factor: **rank by contribution per unit of the scarce resource** (NOT per unit of product); allocate top-rank first up to demand; carve out contractual minimums first.
- Shutdown: continue while **Contribution > Avoidable fixed cost** (net of shutdown/restart costs); shutdown point = contribution equals avoidable fixed cost.
- Two binding constraints → linear programming.

CVP assumptions: linear cost/revenue within relevant range; costs split cleanly fixed/variable; fixed costs constant; production = sales; constant mix; only volume drives cost/revenue (short-run static model).

Worked mini-example

SP ₹50, VC ₹30, Fixed ₹2,00,000, sales 15,000 units. c = 20; P/V = 40%. BEP units = 2,00,000÷20 = 10,000; BEP ₹ = 2,00,000÷0.40 = 5,00,000. MoS = 7,50,000 − 5,00,000 = 2,50,000 (33.33%). Profit = 2,50,000×0.40 = ₹1,00,000. Target ₹1,00,000: units = (2,00,000+1,00,000)÷20 = 15,000. After-tax ₹63,000 @30%: pre-tax = 63,000÷0.70 = 90,000; sales = 2,90,000÷0.40 = ₹7,25,000.

Exam traps & must-remember

1. Full cost lies for decisions — use **contribution**, ignore committed fixed cost.
2. Fixed cost does **not** enter P/V ratio (don't list "reduce fixed cost" to raise P/V).
3. Limiting factor: rank by **contribution per unit of scarce resource**, never per product unit.
4. Absorption vs marginal gap = fixed OH per unit × stock change; separate stock-movement from under/over-absorption.
5. Multi-product "BEP in units" meaningless unless mix fixed — work in sales value.
6. High-low: choose highest/lowest **activity**, watch step/inflation contamination.
7. Special order at full capacity → add opportunity cost of displaced sales.
8. Marginal-cost pricing is a **short-run, spare-capacity** floor only; permanent use = losses.
9. After-tax target must be grossed up before using formulas.
10. Only **avoidable/order-specific** fixed cost is relevant; committed fixed is sunk.
11. Contribution ≠ profit; positive contribution product may still be worth keeping.

One-line recall

- Contribution = Sales − Variable cost = Fixed cost + Profit.

- P/V = Contribution ÷ Sales; BEP = Fixed ÷ (c or P/V); MoS = Actual − BEP; Profit = MoS × P/V.
- Target sales = (Fixed + Target) ÷ P/V; after-tax grossed up by ÷(1−t).
- Rank scarce resource by contribution per unit of that resource.
- Make if buy price > marginal + avoidable fixed + opportunity cost; continue while contribution > avoidable fixed cost.
- Production > Sales → Absorption profit > Marginal; equal only when both methods over the business's life.

Chapter 15 — Budgets & Budgetary Control

Snapshot

A **budget** is a priced, agreed plan that **coordinates** departments, **motivates** with targets, and provides the yardstick for **control** (and delegates authority with control via responsibility accounting). Build starting from the **principal budget factor** (scarcest resource, usually sales). For control you **must flex** the budget to the activity that actually occurred — a fixed budget compared to different actual output mixes volume effect with efficiency and controls nothing. Cash ≠ profit → a separate **cash budget** tracks only real cash flows.

Core concepts

- **Budget** = the quantitative plan; **Budgeting** = preparing it; **Budgetary control** = whole system (prepare → assign → compare actual vs budget → act/revise). Budget ≠ forecast (forecast predicts; budget commits).
- **Principal/key/limiting budget factor**: scarcest resource; build from it. If two candidates, compute output each permits and pick the **smaller**. Common: materials, labour, plant capacity, sales demand, management, finance/cash.
- **Fixed (static) budget** = one activity level (valid for planning/board approval; **wrong for control when activity moves**). **Flexible budget** = recomputed for actual activity.
- Infrastructure: Budget Manual (rulebook), Budget Committee (headed by CEO, reconciles/approves), Budget Officer (coordinates, doesn't impose), Budget Period. Prerequisites: top-management support + responsibility-based structure.
- Classification axes: by **function** (functional vs master), by **flexibility** (fixed vs flexible), by **period** (long/short/current; basic vs current; rolling/continuous).

Key provisions / rules

Functional budget formulas (sequence: sales → production → materials/labour/overheads → cash → master)

- Production budget (units) = Budgeted sales + Desired closing FG stock − Opening FG stock.
- Material **usage** budget = Units produced × material per unit.
- Material **purchase** budget (qty) = Material consumed + Desired closing RM stock − Opening RM stock; value = qty × price.
- Direct labour = units × hours/unit × rate; headcount = total hours ÷ hours per worker.
- Selling & distribution OH often keys off **sales value/volume**, not production.

Flexible budget & high-low

- Total cost at activity x = Total fixed cost + (Variable cost/unit × x).
- High-low variable rate = (Cost at highest activity − Cost at lowest) ÷ (Highest units − Lowest units); pick by **activity**, not cost.
- Fixed part = Total cost at a level − (Variable rate × units at that level).
- Semi-variable: split before flexing; **stepped fixed** cost jumps at band boundary (switch level).
- Variance (control) = Flexed budget − Actual (mark F/A). Volume effect = Flexed budget − Fixed budget (non-controllable).

Capacity/efficiency/activity ratios

- Efficiency = Std hrs for actual output ÷ Actual hrs worked × 100.
- Capacity (utilisation) = Actual hrs worked ÷ Budgeted hrs × 100.
- Activity (volume) = Std hrs for actual output ÷ Budgeted hrs × 100 = **Efficiency × Capacity**.
- Calendar ratio = Actual working days ÷ Budgeted days × 100. Above 100% = favourable.

Cash budget (receipts-and-payments method)

- Closing balance = Opening + Total receipts − Total payments; closing chains to next opening.
- **Exclude** depreciation and all non-cash/accrued items; time flows when cash **moves** (sale collected/purchase paid on lag, not when booked).
- Wrinkles: bad debts never collected; cash discount reduces the discount-taking slice (e.g. 50%×(1−2%) = 49%); sum overlapping collection tails per column; advances = receipt when received; tax/dividend/capex in month **paid**; minimum balance → overdraft (round up to clear floor) or investible surplus.
- Other methods to name: adjusted P&L (cash-flow) method; balance-sheet method.

Master budget = consolidation of all functional budgets (budgeted P&L + balance sheet + cash); operating half + financial half.

ZBB vs Incremental

Aspect	Incremental	Zero-based
Start	Last year + %	Zero
Burden of proof	Only increment	Every rupee
Past waste	Re-funded	Exposed/cut
Cost/time	Low	High
Best for	Stable committed costs	Discretionary/support

ZBB process: define decision units → build decision packages → rank by cost-benefit → fund top-down until funds run out. Related: performance budgeting, activity-based budgeting (ABB), Kaizen budgeting, rolling/continuous budgeting.

Behavioural: budget padding/slack; "spend it or lose it"; dysfunctional behaviour; participative (commitment but slack) vs imposed (tight but demotivating) → negotiated best; aspiration level (tight yet attainable). Feedback (correct after) vs feed-forward (cash budget warns ahead).

Worked mini-example

Overhead at 60% = 12,000 u; Power ₹84,000 (12,000 u) and ₹1,00,000 (16,000 u). High-low: variable = (1,00,000−84,000)÷(16,000−12,000) = ₹4/unit; fixed = 84,000 − 4×12,000 = ₹36,000. Flex power to 100% (20,000 u) = 36,000 + 4×20,000 = ₹1,16,000. Cash-budget check: overhead "₹30,000 incl. ₹8,000 depreciation" → enter cash ₹22,000 only; machinery bought May paid June = June payment.

Exam traps & must-remember

1. **Never** compare a fixed budget with actuals at different activity — **flex first**.
2. **Exclude depreciation** (and accrued/non-cash) from cash budget; subtract from "overheads incl. dep".
3. Timing lags: cash when it **moves**, not when booked; asset paid ≠ bought month.
4. Material **usage ≠ purchases** (differ by RM stock change); two separate reconciliations (FG and RM).
5. Production budget: add closing, subtract opening FG stock; if closing = fraction of **next** period's sales, look forward.
6. When flexing: fixed stays fixed, variable scales, semi-variable split first.
7. Watch **stepped fixed cost** crossing a band — switch fixed figure, don't copy across columns.
8. Label variances F/A and prove they sum to total.
9. Know budget vs forecast; budget vs budgeting vs budgetary control.
10. Minimum cash balance → comment on overdraft/surplus; round overdraft **up** to clear floor.
11. ZBB re-justifies each activity via packages, not "whole company from scratch".
12. High-low: use highest/lowest **activity**, beware abnormal endpoint / hidden fixed-cost step.
13. Collection: discount on discount-taking slice only (49% not 50%); bad debts never collected; sum overlapping tails.
14. Flex each cost against **its own driver** ("2% of sales" keys off sales value, not units).
15. Two constraints → pick the one permitting **less** output as limiting factor.

One-line recall

- Build from the principal budget factor (scarcest resource, usually sales).
- Production (units) = Sales + Closing FG − Opening FG; Purchases = Consumption + Closing RM − Opening RM.
- Flexible budget total = Fixed + (Variable/unit × activity); flex each cost to its driver.
- Control variance = Flexed budget − Actual; volume effect = Flexed − Fixed budget.
- Activity ratio = Efficiency × Capacity; above 100% = favourable.
- Cash budget: Opening + Receipts − Payments = Closing; exclude depreciation, time on cash movement.